A Study on the Foundations of Quantum Field Theory

M.Sc. Project Report

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Abstract

In this report, I have explored the conceptual and mathematical foundations of Quantum Field Theory. In broader sense, QFT is a theory the unification of special relativity and the quantum mechanics through the framework of concepts of fields and their quantization. This report covers the mathematical formulations of free scalar and Dirac fields,and the role of s-matrix and Feynman Diagrams in explaining the interactions of particles. "The study further emphasizes the physical interpretation of field quantization and the use of Feynman diagrams in computing transition amplitudes for various particles' scattering and interactions phenomena.

Keywords: Quantum Field Theory, Scalar Field, Canonical Quantization, Feynman Diagrams, S-Matrix, Dirac Fields.

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Chapter 1

Classical Field Theory

"The mathematical setup of classical field theory is introduced here as a preliminary step towards understanding how quantization and symmetry principles arise to explain the various conservation laws."

1.1 What is Field classically?

According to classical mechanics,a **field** is a physical quantity that takes a definite value at every point in space and time.

Mathematical Definition :

Mathematically, a field is defined as a function of the form :

$$\phi : \mathbb{R}^{3,1} \longrightarrow \mathbb{R} \text{ or } \mathbb{C}$$

that is,

$$\phi = \phi(\mathbf{x}, t),$$

Hence, field $\phi(\mathbf{x}, t)$ assigns a specific value (which may be a scalar, vector, or tensor) to each point in spacetime.

2. Examples

- **Scalar field:** $\phi(\mathbf{x}, t)$ (e.g., temperature, potential)
- **Vector field:** $\mathbf{A}(\mathbf{x}, t)$ (e.g., electric or magnetic field)
- **Tensor field:** $T_{\mu\nu}(\mathbf{x}, t)$ (e.g., metric field $g_{\mu\nu}$ in General Relativity)

Since space is continuous, there are infinitely many points $\mathbf{x}$. Hence, a field has an infinite number of degress of freedom.

1.1.1 Lagrangian Formulation Of Classical Fields

The behavior of a field is described through its **Lagrangian density**, denoted by $\mathcal{L}$, which depends on the field $\phi(\vec{x}, t)$ and its spacetime derivatives $\partial_\mu \phi(\vec{x}, t)$. The **action** for the field is defined by:

$$\boxed{S = \int d^4x \, \mathcal{L}(\phi, \partial_\mu \phi)} \tag{1.1}$$

According to the **principle of least action**, a physical system evolves between two specified states, corresponding to times t_1 and t_2, such that the action S becomes stationary (an extremum), that is,

$$\begin{aligned} 0 &= \delta S \\ &= \int d^4x \left\{ \frac{\partial \mathcal{L}}{\partial \phi} \delta\phi + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \delta(\partial_\mu \phi) \right\} \\ &= \int d^4x \left\{ \frac{\partial \mathcal{L}}{\partial \phi} \delta\phi - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right) \delta\phi + \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \delta\phi \right) \right\} \end{aligned} \tag{1.2}$$

The last term of equation (1.2) can be evaulated as ,

$$\int d^4x \, \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \delta\phi \right) = \oint_{\text{boundary}} dS_\mu \, \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \delta\phi$$

The final term represents a total derivative and thus disappears when $\delta\phi(\vec{x}, t)$ vanishes at spatial infinity and satisfies the boundary conditions $\delta\phi(\vec{x}, t_1) = \delta\phi(\vec{x}, t_2) = 0$. Hence,

$$\delta\phi = 0 \quad \text{on the boundary}$$

which implies:

$$\oint_{\text{boundary}} dS_\mu \, \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \delta\phi = 0$$

Therefore ,equation (1.2) becomes:

$$0 = \int d^4x \left\{ \frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right) \right\} \delta\phi \tag{1.3}$$

Since $\delta\phi$ can take arbitrary values, the expression inside the curly brackets must vanish identically at each spacetime point. Hence, we obtain the **Euler–Lagrange equation** describing the field's equation of motion:

$$\boxed{\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right) - \frac{\partial \mathcal{L}}{\partial \phi} = 0} \tag{1.4}$$

1.1.2 Hamiltonian Formulations of Classical Fields

We know that for a discrete system of particles, conjugate momentum is $p \equiv \partial L / \partial \dot{q}$ for each dynamical variable q and $H = p\dot{q} - L$. So,

$$H = \sum_{\mathbf{x}} p(\mathbf{x}) \dot{\phi}(\mathbf{x}) - L.$$

For fields, this becomes

$$\boxed{\pi(\mathbf{x}) \equiv \frac{\partial \mathcal{L}}{\partial \dot{\phi}(\mathbf{x})}} \tag{1.5}$$

is called the *momentum density* conjugate to $\phi(\mathbf{x})$,and the Hamiltonian is

$$\boxed{\mathcal{H} = \pi(\mathbf{x}) \dot{\phi}(\mathbf{x}) - \mathcal{L}} \tag{1.6}$$

where the **Hamiltonian density** is given by

1.2 Noether's Theorem

For every continuous transformation that leaves the action invariant, there exists a corresponding conserved physical quantity.

For describing a dynamical system of fields, the action is given by

$$S = \int d^4x \mathcal{L}$$

where

$$\mathcal{L} = \mathcal{L}(\phi(x), \partial_\mu \phi(x))$$

under a general set of symmetry transformations;

$$x^\mu \longrightarrow x'^\mu$$

$$\phi(x) \longrightarrow \phi'(x')$$

$$\partial_\mu \phi(x) \longrightarrow \partial'_\mu \phi'(x')$$

If the action **S** remains invariant under these transformations ,then we call them **symmetry** of the physical system.

Key Concept

Every Conservation law arises from certain Continuous symmetry .

There are two types of symmetry transformations:

1. Internal Symmetry Transformation

In this transformation,

$$x^\mu = x'^\mu$$

and only the dynamical variable field $\phi(x)$ varies;

$$\phi(x) \longrightarrow \phi'(x) \quad \text{and} \quad \partial_\mu \phi(x) \longrightarrow \partial_\mu \phi'(x)$$

Now consider the transformation ;

$$\phi(x) \to \phi'(x) = \phi(x) + \alpha \Delta\phi(x),$$

In this expression, α is an infinitesimal quantity characterizing the transformation, while $\Delta\phi$ corresponds to the resulting infinitesimal change in the field configuration. The Lagrangian is transformed as

$$\mathcal{L}(x) \longrightarrow \mathcal{L}'(x) = \mathcal{L}(x) + \alpha \partial_\mu \mathcal{J}^\mu(x), \tag{1.7}$$

for some function $\mathcal{J}^\mu(x)$. We compare this expression for $\Delta\mathcal{L}$ to the one obtained by varying the fields:

$$\alpha \Delta\mathcal{L} = \frac{\partial \mathcal{L}}{\partial \phi}(\alpha \Delta\phi) + \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)}\right) \partial_\mu(\alpha \Delta\phi)$$

$$= \alpha \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \Delta\phi\right) + \alpha \left[\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)}\right)\right] \Delta\phi.$$

The second term vanishes when using the Euler-Lagrange equation (1.4). So we set the remaining term equal to $\alpha \partial_\mu \mathcal{J}^\mu$ and find

$$\boxed{\partial_\mu j^\mu(x) = 0}, \qquad \text{for} \quad \boxed{j^\mu(x) = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \Delta\phi - \mathcal{J}^\mu}. \tag{1.8}$$

Hence, the quantity $j^\mu(x)$ remains conserved and is referred to as the **conserved current**. Each continuous symmetry of the Lagrangian $\mathcal{L}$ gives rise to a corresponding conservation law.

The **conserved charge** is given by

$$\boxed{Q = \int\limits_{\text{all space}} d^3\vec{x}\, j^0(x)} \tag{1.9}$$

which is a constant in time:

$$\frac{d}{dt}Q = 0$$

1.2.1 Derivation of Conserved Current for a Complex Scalar Field

The Lagrangian for a complex scalar field is:

$$\mathcal{L} = |\partial_\mu \phi|^2 - m^2|\phi|^2 = (\partial_\mu \phi)(\partial^\mu \phi^*) - m^2 \phi^* \phi$$

This Lagrangian $\mathcal{L}$ is invariant under the global $U(1)$ transformation:

$$\phi \to e^{i\alpha}\phi, \quad \phi^* \to e^{-i\alpha}\phi^*$$

For the infinitesimal parameter α, the field variations are:

$$\Delta\phi = i\alpha\phi, \quad \Delta\phi^* = -i\alpha\phi^*$$

The Noether current is given by:

$$j^\mu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)}\Delta\phi + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^*)}\Delta\phi^* - \mathcal{J}^\mu$$

where $\Delta\mathcal{L} = \alpha\partial_\mu \mathcal{J}^\mu$. Since $\Delta\mathcal{L} = 0$ for internal transformation, we have $\mathcal{J}^\mu = 0$.

The Lagrangian derivatives are:

$$\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} = \partial^\mu \phi^*, \quad \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^*)} = \partial^\mu \phi$$

Substituting into the current expression:

$$j^\mu = (\partial^\mu \phi^*)(i\alpha\phi) + (\partial^\mu \phi)(-i\alpha\phi^*) = i\alpha\left[(\partial^\mu \phi^*)\phi - \phi^*(\partial^\mu \phi)\right]$$

Removing the infinitesimal parameter α,so the conserved current is:

$$\boxed{j^\mu = i\left[(\partial^\mu \phi^*)\phi - \phi^*(\partial^\mu \phi)\right]}$$

This current satisfies $\partial_\mu j^\mu = 0$, as required by Noether's theorem.

2. Spactime Transformations

For the infinitesimal translation of spacetime coordinate,

$$x^\mu \to x^\mu + a^\mu$$

therefore,

$$\phi(x) \to \phi(x+a) = \phi(x) + a^\mu \partial_\mu \phi(x).$$

Now to find corresponding variation in Lagrangian,

$$\mathcal{L}(x) \longrightarrow \mathcal{L}'(x')$$

Substituting $x' = x + a^\mu$ and expanding to first order by Taylor expansion in a^μ:

$$\mathcal{L}'(x') = \mathcal{L}(x + a^\mu) \approx \mathcal{L}(x) + a^\mu \partial_\mu \mathcal{L}(x)$$

so we get:

$$\mathcal{L}(x) \to \mathcal{L}(x) + a^\mu \partial_\mu \mathcal{L}(x)$$

We can rewrite the variation term $a^\mu \partial_\mu \mathcal{L}(x)$ by introducing the Kronecker delta δ^μ_ν:

$$a^\mu \partial_\mu \mathcal{L}(x) = a^\nu \delta^\mu_\nu \partial_\mu \mathcal{L}(x)$$

$$a^\nu \delta^\mu_\nu \partial_\mu \mathcal{L}(x) = a^\nu \partial_\mu (\delta^\mu_\nu \mathcal{L}(x))$$

therefore ,

$$\mathcal{L}(x) \to \mathcal{L}(x) + a^\nu \partial_\mu (\delta^\mu_\nu \mathcal{L}(x))$$

from equation (1.7) ,

$$\mathcal{L} \to \mathcal{L} + \alpha \partial_\mu \mathcal{J}^\mu$$

we identify $\alpha = a^\nu$ and:

$$\mathcal{J}^\mu = \delta^\mu_\nu \mathcal{L}$$

This nonzero $\mathcal{J}^\mu$ accounts for the explicit spacetime dependence introduced by the coordinate transformation.

Derivation of the Stress-Energy Tensor

We know that the field $\phi(x)$ transforms as:

$$\phi(x) \to \phi(x+a) = \phi(x) + a^\mu \partial_\mu \phi(x)$$

Thus, the variation of the field is:

$$\Delta\phi = \partial_\mu \phi$$

From the Noether's theorem, we know that the conserved current is:

$$j^\mu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \Delta\phi - \mathcal{J}^\mu$$

Substituting our expressions:

$$(j^\mu)_\nu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \partial_\nu \phi - \delta^\mu_{\ \nu} \mathcal{L}$$

Since there are four independent translation directions (one for each component of a^ν), we obtain four conserved currents indexed by ν:

$$\boxed{T^\mu_{\ \nu} \equiv \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \partial_\nu \phi - \mathcal{L}\delta^\mu_{\ \nu}} \tag{1.10}$$

where

$$\mu, \nu = 0, 1, 2, 3$$

which is the **stress-energy tensor** , also called the *energy-momentum tensor*, of the field $\phi(x)$.

The conservation law is $\partial_\mu T^\mu_{\ \nu} = 0$ for each ν.

The **conserved charge** associated with time translations is the *Hamiltonian*:

$$\boxed{H = \int T^{00}\, d^3x = \int \mathcal{H}\, d^3x.} \tag{1.11}$$

The conserved charges associated with spatial translations are

$$\boxed{P^i = \int T^{0i}\, d^3x} \tag{1.12}$$

It represents the **physical momentum** associated with the field, which differs from the canonical momentum.

Chapter–1 Summary

In this chapter, the framework of classical field theory was formulated through both the Lagrangian and Hamiltonian approaches. Starting from a Lagrangian density $\mathcal{L}(\phi, \partial_\mu \phi)$, we obtained the Euler–Lagrange equation which determines the classical

evolution of fields. The conjugate momentum was defined as

$$\pi(x) \equiv \frac{\partial \mathcal{L}}{\partial(\partial_0 \phi)},$$

and led us to the Hamiltonian density

$$\mathcal{H} = \pi\dot{\phi} - \mathcal{L}.$$

We also demonstrated how continuous symmetries of the action produce conserved quantities, expressed in a covariant form through Noether's current

$$\partial_\mu j^\mu(x) = 0.$$

This establishes a direct bridge between symmetry and conservation laws in field theory.

These developments lay the mathematical foundation for field quantization, where fields are promoted to operator-valued distributions and canonical Poisson brackets are replaced by commutators. In subsequent chapters, this machinery will be applied to specific field theories, allowing us to construct propagators, define particle states, and ultimately describe scattering amplitudes in a fully quantum framework.

Chapter 2

Scalar Field Theory

2.1 Scalar Field: Definition

A **scalar field** $\phi(x)$ is a function that assigns a single scalar value to every point x in spacetime.

The properties of Scalar field are :

- **Lorentz Scalar**: Under Lorentz transformations $x^\mu \to x'^\mu = \Lambda^\mu{}_\nu x^\nu$,

$$\phi'(x') = \phi(x)$$

- **Hermiticity**: $\phi(x)^\dagger = \phi(x)$

Examples: Complex Scalar Field, Higgs field in particle physics, Klein-Gordon field.

2.2 Klein-Gordon Equation and Field Quantization

For the scalar field $\phi(x)$, the Lagrangian density is given by

$$\boxed{\mathcal{L} = \frac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) - \frac{1}{2}m^2\phi^2} \tag{2.2}$$

To obtain the equation of motion, the **Euler–Lagrange equation** is applied, yielding:

$$\frac{\partial}{\partial\phi}\mathcal{L} = -m^2\phi$$

$$\frac{\partial}{\partial(\partial_\mu\phi)}\mathcal{L} = \partial^\mu\phi$$

So the equation of motion is:

$$\partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\right) - \frac{\partial\mathcal{L}}{\partial\phi} = \partial_\mu\partial^\mu\phi + m^2\phi = 0$$

$$\partial_\mu \partial^\mu \phi + m^2 \phi = 0$$

or by using d'Alembertian operator $\Box = \partial_\mu \partial^\mu$:

$$\boxed{(\Box + m^2)\phi = 0} \tag{2.3}$$

This is the **Klein-Gordon equation** for the scalar field.

The conjugate momentum is: $\pi(x) = \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = \dot{\phi}(x)$

2.2.1 Solutions of Klien-Gordon equation as a Field Decomposition

Now we have the Klein-Gordon equation ,

$$(\Box + m^2)\phi(x) = 0 \tag{2.4}$$

the solution $\phi(x)$ can be expressed in momentum space by Fourier transform as

$$\phi(x) = \frac{1}{(2\pi)^3} \int \mathrm{d}^4 p \, e^{-ip\cdot x} \widetilde{\phi}(p) \tag{2.5}$$

using the field expansion (2.5) in the Klein-Gordon equation, we obtain

$$\frac{1}{(2\pi)^3} \left(\partial_\mu \partial^\mu + m^2\right) \int \mathrm{d}^4 p \, e^{-ip\cdot x} \widetilde{\phi}(p) = 0,$$

$$\text{or,} \qquad \frac{1}{(2\pi)^3} \int \mathrm{d}^4 p \left(-p^2 + m^2\right) \widetilde{\phi}(p) e^{-ip\cdot x} = 0.$$

So, unless $p^2 = m^2$, $\widetilde{\phi}(p) = 0$.Therefore, it requires **on-shell mass condition**

$$p^2 = (p^0)^2 - \mathbf{p}^2 = m^2. \tag{2.6}$$

$$p^0 = \pm\sqrt{\mathbf{p}^2 + m^2} \equiv \pm E_p.$$

So,we can write: $\widetilde{\phi}(p) = \delta\left(p^2 - m^2\right) a(p)$,

where the operator $a(p)$ is a function at this stage. Substituting this back into the expansion (2.5), we obtain

$$\phi(x) = \frac{1}{(2\pi)^3} \int \mathrm{d}^4 p \, \delta\left(p^2 - m^2\right) e^{-ip\cdot x} a(p). \tag{2.7}$$

by using identity,we have

$$\delta\left(p^2-m^2\right)=\delta\left((p^0)^2-E_p^2\right)=\frac{1}{2E_p}\left[\delta\left(p^0-E_p\right)+\delta\left(p^0+E_p\right)\right].$$

Using this relation in the field expansion (2.7),

$$\phi(x)=\frac{1}{(2\pi)^3}\int dp^0 d^3p\frac{1}{2E_p}\left[\delta\left(p^0-E_p\right)+\delta\left(p^0+E_p\right)\right]\times e^{-ip^0x^0+i\mathbf{p}\cdot\mathbf{x}}a(p^0,\mathbf{p})$$

$$=\frac{1}{(2\pi)^3}\int d^3p\frac{1}{2E_p}\left(e^{-iE_px^0+i\mathbf{p}\cdot\mathbf{x}}a(E_p,\mathbf{p})+e^{iE_px^0+i\mathbf{p}\cdot\mathbf{x}}a(-E_p,\mathbf{p})\right).$$

Changing $\mathbf{p}\to-\mathbf{p}$ in the second term and identifying $p^0=E_p>0$, we obtain

$$\phi(x)=\frac{1}{(2\pi)^3}\int\frac{d^3p}{2p^0}\left(e^{-ip\cdot x}a(p)+e^{ip\cdot x}a(-p)\right).$$

so that we can write

$$\phi(x)=\frac{1}{(2\pi)^3}\int\frac{d^3p}{2p^0}\left(e^{-ip\cdot x}a(p)+e^{ip\cdot x}a^\dagger(p)\right). \tag{2.8}$$

where, $\quad a^\dagger(p)=a(-p),\quad a^\dagger(-p)=a(p),$

By defining

$$a_{\mathbf{p}}=\frac{a(p)}{\sqrt{2p^0}},\quad a^\dagger_{\mathbf{p}}=\frac{a^\dagger(p)}{\sqrt{2p^0}},$$

we can write the field operator (2.8) as

$$\boxed{\phi(x)=\int\frac{d^3p}{(2\pi)^3\sqrt{2p^0}}\left(e^{-ip\cdot x}\,a(\mathbf{p})+e^{ip\cdot x}\,a^\dagger(\mathbf{p})\right).} \tag{2.9}$$

We know that conjugate momentum for the scalar field is given by $\pi=\dot{\phi(x)}$,so

$$\boxed{\pi(\mathbf{x})=\int\frac{d^3p}{(2\pi)^3}(-i)\sqrt{\frac{p^0}{2}}\left(a_{\mathbf{p}}e^{i\mathbf{p}\cdot\mathbf{x}}-a^\dagger_{\mathbf{p}}e^{-i\mathbf{p}\cdot\mathbf{x}}\right)} \tag{2.10}$$

2.2.2 Green's Function and Feynman's Propagator

Consider a Klein–Gordon field interacting with an external source $J(x)$ that induces the creation or annihilation of field quanta described by $\phi(x)$. The Lagrangian for this

system takes the form :

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi)(\partial^\mu \phi) - \frac{1}{2}m^2\phi^2 + J(x)\phi(x) \tag{2.11}$$

Now,

$$\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} = \partial^\mu \phi, \quad \frac{\partial \mathcal{L}}{\partial \phi} = -m^2\phi + J(x)$$

Substitute into the Euler-Lagrange equation (1.4)

$$\partial_\mu(\partial^\mu \phi) - (-m^2\phi + J(x)) = 0$$

$$\partial_\mu \partial^\mu \phi + m^2\phi - J(x) = 0$$

So the equation of motion is:

$$\boxed{(\Box + m^2)\phi(x) = J(x)} \tag{2.12}$$

Green's Function

The Green's function $G(x-y)$ satisfies equation (2.12) with a delta function source as:

$$(\Box + m^2)G(x-y) = -i\delta^4(x-y) \tag{2.13}$$

the solution is written as

$$\phi(x) = -\int d^4x G(x-y)J(y)$$

Now, by using Fourier transform, we can write:

$$\delta^4(x-y) = \frac{1}{(2\pi)^4}\int d^4p\, e^{-ip\cdot(x-y)} \tag{2.14}$$

$$G(x-y) = \int \frac{d^4p}{(2\pi)^4}\, e^{-ip\cdot(x-y)}\, \widetilde{G}(p) \tag{2.15}$$

By substitution of (2.14) and (2.15) in (2.13), we get:

$$(\Box + m^2)\int \frac{d^4p}{(2\pi)^4} e^{ip\cdot(x-y)}\widetilde{G}(p) = -i\int \frac{d^4p}{(2\pi)^4} e^{ip\cdot(x-y)}$$

Thus:

$$\widetilde{G}(p) = \frac{i}{p^2 - m^2} \tag{2.16}$$

Now put this back into equation (2.15), we get:

$$G(x-y) = \int \frac{d^4p}{(2\pi)^4} e^{ip\cdot(x-y)} \frac{i}{p^2 - m^2} \tag{2.17}$$

This is the Green's function for field $\phi(x)$ in presence of source $J(x)$. The poles of the integrand exist at

$$p^2 = m^2 \quad \text{or} \quad (p^0)^2 - \vec{p}^2 = m^2 \text{or} \quad p^0 = \pm\sqrt{\vec{p}^2 + m^2}$$

which lie on the real axis of the complex p^0 plane. In order to solve this integral for retarded $(x^0 - y^0 > 0)$ or advanced $(x^0 - y^0 < 0)$ times, we shift the poles either upward or downward from the real axis by introducing Feynman $i\epsilon$ prescription,so that the poles are then at $p^0 = \pm(E_p - i\epsilon)$ and so the Green's function (2.17) can be written as:

$$\boxed{G_F(x-y) = \int \frac{d^4p}{(2\pi)^4} \frac{ie^{ip\cdot(x-y)}}{p^2 - m^2 + i\epsilon}} \tag{2.18}$$

This is the **Feynman's propagator** .

Case (1):

For $x^0 - y^0 < 0$, we will close the contour in the upper half plane for the damping of the exponential and so this will pick up the residue of the pole at $p^0 = -E_p + i\epsilon$. Therefore, for $x^0 - y^0 < 0$, we have:

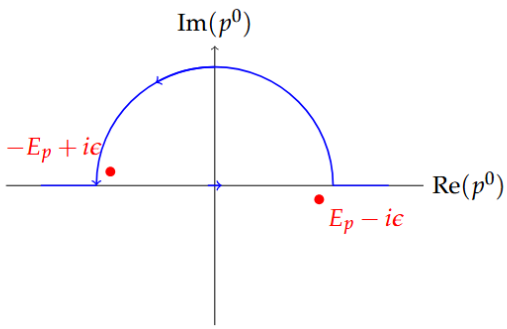


Figure 2.1: Contour for $x^0 - y^0 < 0$ — closed in upper half plane (counterclockwise)

$$\begin{aligned} G_F(x-y) &= \lim_{\epsilon\to 0^+} \int \frac{d^3p}{(2\pi)^4} \int dp^0 \frac{ie^{-ip\cdot(x-y)}}{(p^0 - E_p + i\epsilon)(p^0 + E_p - i\epsilon)} \\ &= \lim_{\epsilon\to 0^+} 2\pi i \int \frac{d^3p}{(2\pi)^4} \frac{ie^{-i(-E_p+i\epsilon)(x^0-y^0)+i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})}}{2(-E_p + i\epsilon)} \end{aligned}$$

$$= \frac{1}{2}\int \frac{d^3p}{(2\pi)^3}\frac{e^{iE_p(x^0-y^0)+i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})}}{E_p}$$

$$= \frac{1}{2}\int \frac{d^3p}{(2\pi)^3}\frac{e^{ip\cdot(x-y)}}{2E_p}$$

where x-y has been replaced by x for convenience. This is called the **advanced Green's function**.

Case (2):

For $x^0 - y^0 > 0$, we close the contour in the lower half plane for the damping of the exponential and this will pick up the residue of the pole at $p^0 = E_p - i\epsilon$. Therefore, for $x^0 - y^0 > 0$, we have:

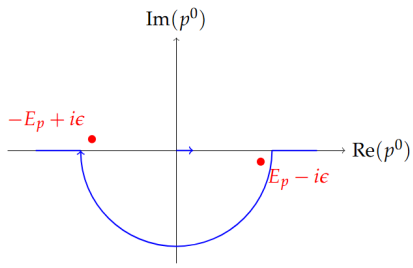


Figure 2.2: Contour for $x^0 - y^0 > 0$ — closed in lower half plane (clockwise)

$$G_F(x-y) = \lim_{\epsilon\to 0^+}\int \frac{d^3p}{(2\pi)^4}\int dp^0 \frac{ie^{-ip\cdot(x-y)}}{(p^0 - E_p + i\epsilon)(p^0 + E_p - i\epsilon)}$$

$$= \lim_{\epsilon\to 0^+}(-2\pi i)\int \frac{d^3p}{(2\pi)^4}\frac{ie^{-i(E_p-i\epsilon)(x^0-y^0)+i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})}}{2(E_p - i\epsilon)}$$

$$= -\frac{1}{2}\int \frac{d^3p}{(2\pi)^3}\frac{e^{-iE_p(x^0-y^0)+i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})}}{E_p}$$

$$= -\frac{1}{2}\int \frac{d^3p}{(2\pi)^3}\frac{e^{-ip\cdot(x-y)}}{2E_p}$$

where x-y has been replaced by x for convenience.

We get negative sign due to clockwise contour integration in lower half plan.

This is called the **retarded Green's function**.

Thus, we have

$$G_F(x-y) = \begin{cases} G(x-y) & \text{for } x^0 > y^0 \\ G(y-x) & \text{for } x^0 < y^0 \end{cases}$$

$$= \theta(x^0 - y^0)\langle 0|\phi(x)\phi(y)|0\rangle + \theta(y^0 - x^0)\langle 0|\phi(y)\phi(x)|0\rangle$$

$$\equiv \langle 0|T\phi(x)\phi(y)|0\rangle.$$

where T is the **time ordering** symbol, defined as follow:

$$T\phi(x)\phi(y) = \begin{cases} \phi(x)\phi(y) & \text{for } x^0 > y^0 \\ \phi(y)\phi(x) & \text{for } x^0 < y^0 \end{cases}$$

2.3 Commutation Relations

To derive commutation relations for fields $\phi(x)$ and $\phi(y)$, we use the following commutation relations between creation and annihilation operators:

$$[a(\mathbf{k}), a(\mathbf{k}')] = 0 = [a^\dagger(\mathbf{k}), a^\dagger(\mathbf{k}')],$$

$$[a(\mathbf{k}), a^\dagger(\mathbf{k}')] = \delta^3(\mathbf{k} - \mathbf{k}').$$

Derivation of Equal Time Commutation Relation

We know that the field expansions are :

$$\phi(x) = \int \frac{d^3p}{\sqrt{(2\pi)^3 2E_p}} \left(a(\mathbf{p})e^{-ip\cdot x} + a^\dagger(\mathbf{p})e^{ip\cdot x}\right)$$

$$\pi(y) = \dot{\phi}(y) = \int \frac{d^3p'}{\sqrt{(2\pi)^3 2E_{p'}}}(-iE_{p'}) \left(a(\mathbf{p}')e^{-ip'\cdot y} - a^\dagger(\mathbf{p}')e^{ip'\cdot y}\right)$$

We compute the commutator at equal time $x^0 = y^0$:

$$[\phi(x), \pi(y)]_{x^0=y^0} = \phi(x)\pi(y) - \pi(y)\phi(x)$$

By substitute the field expansions:

$$[\phi(x), \pi(y)] = \int \frac{d^3p d^3p'}{(2\pi)^3\sqrt{4E_pE_{p'}}}(-iE_{p'})\times$$

$$\left[a(\mathbf{p})a(\mathbf{p}')e^{-ip\cdot x}e^{-ip'\cdot y} - a(\mathbf{p})a^\dagger(\mathbf{p}')e^{-ip\cdot x}e^{ip'\cdot y} + a^\dagger(\mathbf{p})a(\mathbf{p}')e^{ip\cdot x}e^{-ip'\cdot y} - a^\dagger(\mathbf{p})a^\dagger(\mathbf{p}')e^{ip\cdot x}e^{ip'\cdot y}\right]$$

$$-\left[a(\mathbf{p}')a(\mathbf{p})e^{-ip'\cdot y}e^{-ip\cdot x} - a^\dagger(\mathbf{p}')a(\mathbf{p})e^{ip'\cdot y}e^{-ip\cdot x} + a(\mathbf{p}')a^\dagger(\mathbf{p})e^{-ip'\cdot y}e^{ip\cdot x} - a^\dagger(\mathbf{p}')a^\dagger(\mathbf{p})e^{ip'\cdot y}e^{ip\cdot x}\right]$$

Using the commutation relations:

$$[a(\mathbf{p}), a(\mathbf{p}')] = 0, \quad [a^\dagger(\mathbf{p}), a^\dagger(\mathbf{p}')] = 0, \quad [a(\mathbf{p}), a^\dagger(\mathbf{p}')] = \delta^3(\mathbf{p} - \mathbf{p}')$$

The non-zero terms come from:

$$a(\mathbf{p})a^\dagger(\mathbf{p}') - a^\dagger(\mathbf{p}')a(\mathbf{p}) = \delta^3(\mathbf{p} - \mathbf{p}')$$

$$a^\dagger(\mathbf{p})a(\mathbf{p}') - a(\mathbf{p}')a^\dagger(\mathbf{p}) = -\delta^3(\mathbf{p} - \mathbf{p}')$$

By simplification and using $x^0 = y^0$, we get:

$$[\phi(x), \pi(y)] = i\int \frac{d^3p}{(2\pi)^3} e^{i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})}$$

since the Dirac delta function is ,

$$\int \frac{d^3p}{(2\pi)^3} e^{i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} = \delta^3(\mathbf{x} - \mathbf{y})$$

The

$$\boxed{[\phi(\mathbf{x}, t), \pi(\mathbf{y}, t)] = i\delta^3(\mathbf{x} - \mathbf{y})}$$

Similarly,we get following commutation relations:

$$[\phi(\mathbf{x}, t), \phi(\mathbf{y}, t)] = 0 \quad \text{and} \quad [\pi(\mathbf{x}, t), \pi(\mathbf{y}, t)] = 0$$

2.4 Quantized Hamiltonian For The Scalar Field

We know that $\mathcal{H} = \pi\dot{\phi} - \mathcal{L}$

For scalar field: $\mathcal{L} = \frac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) - \frac{1}{2}m^2\phi^2$ and $\pi = \dot{\phi}$

Substituting L,

$$\mathcal{H} = \pi\dot{\phi} - \left(\frac{1}{2}\dot{\phi}^2 - \frac{1}{2}(\nabla\phi)^2 - \frac{1}{2}m^2\phi^2\right)$$

The total Hamiltonian is:

$$H = \int d^3x\, \mathcal{H} = \int d^3x \left[\frac{1}{2}\pi^2 + \frac{1}{2}\nabla\phi.\nabla\phi + \frac{1}{2}m^2\phi^2\right] \tag{2.19}$$

Now we evaluate each term of the Hamiltonian by using mode expansions of $\phi(x) and \pi(x)$:

1.
$$\int \pi^2(x)d^3x = \int d^3x \int \frac{d^3p d^3p'}{(2\pi)^3\sqrt{4E_pE_{p'}}}(-iE_p)(-iE_{p'})\times$$
$$\left(a(\mathbf{p})e^{-ip\cdot x} - a^\dagger(\mathbf{p})e^{ip\cdot x}\right)\left(a(\mathbf{p}')e^{-ip'\cdot x} - a^\dagger(\mathbf{p}')e^{ip'\cdot x}\right)$$

Using $\int d^3x e^{i(\mathbf{p}\pm\mathbf{p}')\cdot\mathbf{x}} = (2\pi)^3\delta^3(\mathbf{p}\pm\mathbf{p}')$ and simplifying:

$$\int \pi^2(x)d^3x = \frac{1}{2}\int d^3pE_p\left(a(\mathbf{p}) - a^\dagger(-\mathbf{p})\right)\left(a(-\mathbf{p}) - a^\dagger(\mathbf{p})\right)$$

2. $\int(\nabla\phi)^2d^3x = \int d^3x\,(\nabla\phi(\mathbf{x})\cdot\nabla\phi(\mathbf{x}))$

$$\nabla\phi(x) = \int \frac{d^3p}{(2\pi)^3\sqrt{2E_p}}(i\mathbf{p})\left(a(\mathbf{p})e^{-ip\cdot x} - a^\dagger(\mathbf{p})e^{ip\cdot x}\right)$$

by similar steps

$$\int(\nabla\phi)^2d^3x = \frac{1}{2}\int d^3p\frac{\mathbf{p}^2}{E_p}\left(a(\mathbf{p}) - a^\dagger(-\mathbf{p})\right)\left((a(-\mathbf{p}) + a^\dagger(\mathbf{p})\right)$$

3. $\int m^2\phi^2d^3x = \int d^3xm^2\phi(\mathbf{x})\phi(\mathbf{x})$

$$\int m^2\phi^2d^3x = \frac{1}{2}\int d^3p\frac{m^2}{E_p}\left(a(\mathbf{p}) + a^\dagger(-\mathbf{p})\right)\left(a(-\mathbf{p}) + a^\dagger(\mathbf{p})\right)$$

Combining all three terms and using commutation relations of operators ,we get:

$$H = \frac{1}{2}\int d^3p\left[E_p + \frac{\mathbf{p}^2}{E_p} + \frac{m^2}{E_p}\right]\left(a(\mathbf{p})a^\dagger(\mathbf{p}) + a^\dagger(\mathbf{p})a(\mathbf{p})\right)$$

Since $E_p^2 = \mathbf{p}^2 + m^2$, we have:

$$E_p + \frac{\mathbf{p}^2}{E_p} + \frac{m^2}{E_p} = E_p + \frac{E_p^2}{E_p} = 2E_p$$

Therefore:

$$H = \frac{1}{2}\int d^3p2E_p\left(a(\mathbf{p})a^\dagger(\mathbf{p}) + a^\dagger(\mathbf{p})a(\mathbf{p})\right) = \int d^3pE_p\left(a^\dagger(\mathbf{p})a(\mathbf{p}) + \frac{1}{2}[a(\mathbf{p}), a^\dagger(\mathbf{p})]\right)$$

Using $[a(\mathbf{p}), a^\dagger(\mathbf{p}')] = \delta^3(\mathbf{p}-\mathbf{p}')$:

$$\boxed{H = \int d^3pE_p\left(a^\dagger(\mathbf{p})a(\mathbf{p}) + \frac{1}{2}\delta^3(0)\right)} \tag{2.20}$$

If we use **Normal Ordering** , then it can be written as :

$$\boxed{:H:=\int d^3p\, E_{\vec{p}\, a^\dagger(\mathbf{p})a(\mathbf{p})}} \tag{2.21}$$

where the normal ordering $:H:$ removes the infinite zero-point energy term.

2.4.1 Quantized Momentum

From equation (1.12),

$$P^i = \int T^{0i}\, d^3x$$

The energy-momentum tensor is given by from equation (1.10);

$$T^{\mu\nu} = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\partial^\nu\phi - \eta^{\mu\nu}\mathcal{L}$$

also using

$$\mathcal{L} = \frac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) - \frac{1}{2}m^2\phi^2$$

For T^{0i}:

$$T^{0i} = \frac{\partial\mathcal{L}}{\partial(\partial_0\phi)}\partial^i\phi = \pi\partial^i\phi$$

Therefore:

$$P^i = \int d^3x\, \pi(\mathbf{x})\partial^i\phi(\mathbf{x})$$

Substitute the field expansions:

$$P^i = \int d^3x \int \frac{d^3p d^3p'}{(2\pi)^3\sqrt{4E_pE_{p'}}}(-iE_p)(ip'^i)\times$$

$$\left(a(\mathbf{p})e^{-ip\cdot x} - a^\dagger(\mathbf{p})e^{ip\cdot x}\right)\left(a(\mathbf{p}')e^{-ip'\cdot x} - a^\dagger(\mathbf{p}')e^{ip'\cdot x}\right)$$

Using $\int d^3x e^{i(\mathbf{p}\pm\mathbf{p}')\cdot\mathbf{x}} = (2\pi)^3\delta^3(\mathbf{p}\pm\mathbf{p}')$ and simplifying:

$$P^i = \frac{1}{2}\int d^3p\, p^i\left(a(\mathbf{p})a^\dagger(\mathbf{p}) + a^\dagger(\mathbf{p})a(\mathbf{p})\right)$$

Using normal ordering,we get:

$$\boxed{P^i = \int d^3p\, p^i\, a^\dagger(\mathbf{p})a(\mathbf{p})} \tag{2.22}$$

Chapter–2 Summary

In this chapter, we studied the quantization of a real scalar field. We started with the classical Lagrangian. From this, we obtained the field equation of motion. We then promoted the classical field to an operator by expressing it in terms of Fourier modes. We evaluated the Feynman propagator $G_F(x-y)$ using complex contour integration. The Hamiltonian took the form

$$H = \int d^3p\, E_p\, a^\dagger(\mathbf{p})a(\mathbf{p}),$$

showing that the free scalar field acts mathematically like a group of independent quantum harmonic oscillators.

This setup allowed us to view particles as excitations of the underlying field. The techniques we developed here lay the groundwork for more complex quantum field theories, where interactions between fields will change propagators and add non-trivial dynamics through perturbative corrections.

Chapter 3

Dirac Field Theory

3.1 Introduction

The Dirac field theory provides the correct relativistic quantum description of spin-$\frac{1}{2}$ particles, and eliminates the negative energy problems of the Klein-Gordon equation . Dirac equation is a first-order differential equation that naturally incorporates spin and predicts the existence of antimatter.

3.1.1 Dirac Equation

Dirac proposed a first-order relativistic wave equation which treats space and time on equal footing, unlike the second-order Klein-Gordon equation. The Dirac equation is written as:

$$i\hbar\frac{\partial\psi}{\partial t} = H\psi$$

$$i\hbar\frac{\partial\psi}{\partial t} = \left(-i\hbar c\boldsymbol{\alpha}\cdot\nabla + \beta mc^2\right)\psi$$

where $\alpha^i = \gamma^0\gamma^i$ and $\beta = \gamma^0$.

The **Dirac Hamiltonian** is given by

$$H = -i\hbar c\boldsymbol{\alpha}\cdot\nabla + \beta mc^2 \tag{3.1}$$

Since momentum $p_\mu = i\hbar\partial_\mu$, we can write the Dirac equation as:

$$i\hbar\frac{\partial\psi}{\partial t} = \left(c\boldsymbol{\alpha}\cdot\mathbf{p} + \beta mc^2\right)\psi$$

In covariant form (taking $\hbar = c = 1$):

$$\boxed{(i\gamma^\mu\partial_\mu - m)\psi = 0} \tag{3.2}$$

3.1.2 Dirac Gamma Matrices

The four Dirac gamma matrices γ^μ ($\mu = 0, 1, 2, 3$) are defined by the **Clifford algebra** anticommutation relations:

$$\boxed{\{\gamma^\mu, \gamma^\nu\} = \gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu = 2\eta^{\mu\nu} I_4} \tag{3.3}$$

where the quantity $\eta^{\mu\nu} = \text{diag}(1, -1, -1, -1)$ is the Minkowski metric and I_4 is the 4×4 identity matrix.

Properties:

$$(\gamma^0)^2 = I_4, \quad (\gamma^i)^2 = -I_4 \quad (i = 1, 2, 3), \quad \gamma^\mu\gamma^\nu = -\gamma^\nu\gamma^\mu \quad (\mu \neq \nu)$$

$$\{\gamma^\mu, \gamma^\nu\} = \gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu = 2\eta^{\mu\nu} I$$

Representations

1. Dirac Representation

$$\gamma^0 = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}, \quad \gamma^i = \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix} \tag{3.4}$$

where σ^i are the Pauli matrices:

$$\sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

2. Weyl (Chiral) Representation

$$\gamma^0 = \begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix}, \quad \gamma^i = \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix} \tag{3.5}$$

Fifth Gamma Matrix : The matrix γ^5 is defined as:

$$\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$$

In the Dirac representation:

$$\gamma^5 = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}$$

Properties of γ^5:

$$(\gamma^5)^2 = I_4, \quad \{\gamma^5, \gamma^\mu\} = 0, \quad (\gamma^5)^\dagger = \gamma^5$$

3.2 Free Particle Solutions of Dirac Equation

We can write the wavefunction $\psi(x)$ as:

$$\Psi = u(p,s)e^{-ip\cdot x} \quad \text{for positive energy solution} \tag{3.7}$$

and

$$\Psi = v(p,s)e^{ip\cdot x} \quad \text{for negative energy solution} \tag{3.8}$$

where both $u(p,s)$ and $v(p,s)$ are 4-component spinors, and s denotes the spin. Substituting equation (3.7) in the Dirac equation, we obtain:

$$(i\gamma^\mu \partial_\mu - m)u(p,s)e^{-ip\cdot x} = 0$$

$$(i\gamma^\mu(-ip_\mu) - m)u(p,s)e^{-ip\cdot x} = 0$$

$$(\gamma^\mu p_\mu - m)u(p,s)e^{-ip\cdot x} = 0$$

Since $e^{-ip\cdot x} \neq 0$, we get:

$$(\gamma^\mu p_\mu - m)u(p,s) = 0 \tag{3.9}$$

similarly ,by putting (3.8) in Dirac equation,we get:

$$(\gamma^\mu p_\mu + m)v(p,s) = 0 \tag{3.10}$$

In the Rest Frame

In the rest frame, $\mathbf{p} = 0$, hence from Eq. (3.9):

$$(\gamma^0 p_0 - m)u(p^0,s) = 0$$

Thus,

$$(\gamma^0 m - m)u(p^0,s) = 0 \quad \Rightarrow \quad (\gamma^0 - I_{4\times4})u(p^0,s) = 0 \tag{3.11}$$

Using Weyl representation:

$$\gamma^0 = \begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix}$$

we have:

$$(\gamma^0 - I)u(p^0, s) = \left[\begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix} - \begin{pmatrix} I_2 & 0 \\ 0 & I_2 \end{pmatrix} \right] u(p^0, s) = \begin{pmatrix} -I_2 & I_2 \\ I_2 & -I_2 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{pmatrix}$$

where I_2 is a 2×2 unit matrix:

$$I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Writing $up^0, s) = \begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{pmatrix} = \begin{pmatrix} \xi \\ \eta \end{pmatrix}$ where $\xi = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}, \eta = \begin{pmatrix} u_3 \\ u_4 \end{pmatrix}$; we have now

$$\begin{pmatrix} -I_2 & I_2 \\ I_2 & -I_2 \end{pmatrix} \begin{pmatrix} \xi \\ \eta \end{pmatrix} = 0$$

From this ,we get:

$$-\xi + \eta = 0$$
$$\xi - \eta = 0$$

Both equations give $\xi = \eta$. Therefore, in the rest frame, positive energy solution is:

$$u(p^0, s) = \begin{pmatrix} \xi \\ \xi \end{pmatrix}$$

where ξ is an arbitrary 2-component spinor. Choosing the basis:

$$\xi^{(1)} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \text{ for spin up}, \quad \xi^{(2)} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \text{ for spin down}$$

we obtain the two independent positive energy solutions in the rest frame:

$$u(p^0, 1) = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix} \text{ for spin up}, \quad u(p^0, 2) = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix} \text{ for spin down}$$

For negative energy solutions, from equation (3.10), in the rest frame:

$$(\gamma^0 m + m)v(p^0, s) = 0 \quad \Rightarrow \quad (\gamma^0 + I)v(p^0, s) = 0$$

$$(\gamma^0 + I)v(p^0, s) = \begin{pmatrix} I_2 & I_2 \\ I_2 & I_2 \end{pmatrix} \begin{pmatrix} \xi \\ \eta \end{pmatrix} = 0$$

This gives $\xi + \eta = 0$, so $\eta = -\xi$. Therefore:

$$v(p^0, s) = \begin{pmatrix} \xi \\ -\xi \end{pmatrix}$$

Choosing the same basis as before, we get:

$$v(p^0, 1) = \begin{pmatrix} -1 \\ 0 \\ 1 \\ 0 \end{pmatrix} \text{ for spin up}, \quad v(p^0, 2) = \begin{pmatrix} 0 \\ 1 \\ 0 \\ -1 \end{pmatrix} \text{ for spin down}$$

3.2.1 Relativistic Normalization

To normalize the spinor, we impose the condition $u^\dagger u = 2E_p$. Let the spinor be written as $u = Nu_0$ and $u^\dagger = Nu_0^\dagger$, which leads to $N^2 u_0^\dagger u_0 = 2E_p$. In the particle's rest frame, this yields $N = \sqrt{m}$, where N acts as the normalization constant. Hence, in the rest frame, the spinor takes the form $\zeta = \sqrt{m}\,\xi$. Under a Lorentz boost, these spinors transform according to

$$u(p, s) = Ru(0, s) = \begin{pmatrix} R_L & 0 \\ 0 & R_R \end{pmatrix} \begin{pmatrix} \sqrt{m}\xi \\ \sqrt{m}\xi \end{pmatrix} = \begin{pmatrix} \sqrt{m}R_L\xi \\ \sqrt{m}R_R\xi \end{pmatrix} \tag{3.12}$$

where R_L and R_R are the left-handed and right-handed boost/Lorentz transformation matrices, given by:

$$R_L = \sqrt{\gamma - \beta\gamma\hat{n} \cdot \vec{\sigma}}$$

and

$$R_R = \sqrt{\gamma + \beta\gamma\hat{n} \cdot \vec{\sigma}}$$

So now the solutions for a particle of mass "m" with momentum ,are given by using the result in Eq. (3.11) and using $\gamma = (E/m)$, $\gamma\beta\hat{n} = \mathbf{p}/m$, we find that the boosted spinors have the form

$$u(p,s) = \begin{pmatrix} \sqrt{E - \mathbf{p}\cdot\vec{\sigma}}\,\xi^s \\ \sqrt{E + \mathbf{p}\cdot\vec{\sigma}}\,\xi^s \end{pmatrix}; \quad v(p,s) = \begin{pmatrix} \sqrt{E - \mathbf{p}\cdot\vec{\sigma}}\,\eta^s \\ -\sqrt{E + \mathbf{p}\cdot\vec{\sigma}}\,\eta^s \end{pmatrix} \tag{3.13}$$

where ξ and η are 2-component spinors representing the spin states with s=1 for spin up and s=2 for spin down notations.

3.2.2 Spin Sums and Identities

1. **Identities to be used for spin sums** :

First Identity: $\sqrt{p\cdot\sigma}\sqrt{p\cdot\bar{\sigma}} = m$

$$(p\cdot\sigma)(p\cdot\bar{\sigma}) = (p^0 - \mathbf{p}\cdot\vec{\sigma})(p^0 + \mathbf{p}\cdot\vec{\sigma}) = (p^0)^2 - (\mathbf{p}\cdot\vec{\sigma})^2$$

Using $(\mathbf{p}\cdot\vec{\sigma})^2 = \mathbf{p}^2 I_2$:

$$(p\cdot\sigma)(p\cdot\bar{\sigma}) = (p^0)^2 - \mathbf{p}^2 = p^2 = m^2 I_2$$

Taking square roots:

$$\sqrt{p\cdot\sigma}\sqrt{p\cdot\bar{\sigma}} = \sqrt{(p\cdot\sigma)(p\cdot\bar{\sigma})} = \sqrt{m^2 I_2} = m I_2$$

Second Identity: $\sqrt{p\cdot\sigma}\sqrt{p\cdot\sigma} = p\cdot\sigma$ This follows directly from the definition of the matrix square root:

$$\sqrt{p\cdot\sigma}\sqrt{p\cdot\sigma} = p\cdot\sigma$$

Third Identity: $\sqrt{p\cdot\bar{\sigma}}\sqrt{p\cdot\bar{\sigma}} = p\cdot\bar{\sigma}$ Similarly, by taking square root:

$$\sqrt{p\cdot\bar{\sigma}}\sqrt{p\cdot\bar{\sigma}} = p\cdot\bar{\sigma}$$

2. **Spin Sums**

$$\sum_{s=1,2} u^s(p)\bar{u}^s(p) = \sum_s \begin{pmatrix} \sqrt{p\cdot\sigma}\,\xi^s \\ \sqrt{p\cdot\bar{\sigma}}\,\xi^s \end{pmatrix} \left(\xi^{s\dagger}\sqrt{p\cdot\bar{\sigma}},\ \ \xi^{s\dagger}\sqrt{p\cdot\sigma}\right)$$

$$= \begin{pmatrix} \sqrt{p\cdot\sigma}\sqrt{p\cdot\bar{\sigma}} & \sqrt{p\cdot\sigma}\sqrt{p\cdot\sigma} \\ \sqrt{p\cdot\bar{\sigma}}\sqrt{p\cdot\bar{\sigma}} & \sqrt{p\cdot\bar{\sigma}}\sqrt{p\cdot\sigma} \end{pmatrix}$$

$$= \begin{pmatrix} m & p\cdot\sigma \\ p\cdot\bar{\sigma} & m \end{pmatrix}$$

$$= \gamma\cdot p + m I_{4\times 4}$$

$$\boxed{\sum_{s=1,2} u^s(p)\bar{u}^s(p) = \gamma \cdot p + mI_{4\times 4}} \tag{3.14}$$

3.

Spin sums :

We know that

$$v(p,s) = \begin{pmatrix} \sqrt{p\cdot\sigma}\,\eta^s \\ -\sqrt{p\cdot\bar{\sigma}}\,\eta^s \end{pmatrix}, \quad \bar{v}(p,s) = \begin{pmatrix} \eta^{s\dagger}\sqrt{p\cdot\sigma}, & -\eta^{s\dagger}\sqrt{p\cdot\bar{\sigma}} \end{pmatrix}$$

so now ;

$$\sum_s v^s(p)\bar{v}^s(p) = \sum_s \begin{pmatrix} \sqrt{p\cdot\sigma}\,\eta^s \\ -\sqrt{p\cdot\bar{\sigma}}\,\eta^s \end{pmatrix} \begin{pmatrix} \eta^{s\dagger}\sqrt{p\cdot\sigma}, & -\eta^{s\dagger}\sqrt{p\cdot\bar{\sigma}} \end{pmatrix}$$

$$= \begin{pmatrix} \sqrt{p\cdot\sigma}\sqrt{p\cdot\sigma} & -\sqrt{p\cdot\sigma}\sqrt{p\cdot\bar{\sigma}} \\ -\sqrt{p\cdot\bar{\sigma}}\sqrt{p\cdot\sigma} & \sqrt{p\cdot\bar{\sigma}}\sqrt{p\cdot\bar{\sigma}} \end{pmatrix}$$

Using the identities:

$$\sqrt{p\cdot\sigma}\sqrt{p\cdot\sigma} = p\cdot\sigma, \quad \sqrt{p\cdot\sigma}\sqrt{p\cdot\bar{\sigma}} = m, \quad \sqrt{p\cdot\bar{\sigma}}\sqrt{p\cdot\bar{\sigma}} = p\cdot\bar{\sigma}$$

We get:

$$\boxed{\sum_s v^s(p)\bar{v}^s(p) = \begin{pmatrix} p\cdot\sigma & -m \\ -m & p\cdot\bar{\sigma} \end{pmatrix} = \gamma\cdot p - m} \tag{3.15}$$

4.

completeness relation:

We choose the basis spinors to be:

$$\xi^1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \xi^2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Then:

$$\xi^1\xi^{1\dagger} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$\xi^2\xi^{2\dagger} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

Adding both terms:

$$\sum_{s=1,2} \xi^s \xi^{s\dagger} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I_2 \tag{3.16}$$

This is the **completeness relation**for the 2-component spinor basis.

3.3 Quantization Of Dirac Field

So we Fourier expand the Dirac field $\psi(x)$ in these bases as:

$$\boxed{\psi(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_p}} \sum_{s=1,2} \left[a^s_{\mathbf{p}} u^s(p) e^{-ip\cdot x} + b^{s\dagger}_{\mathbf{p}} v^s(p) e^{ip\cdot x} \right]} \tag{3.17}$$

where:

- $a^s_{\mathbf{p}}$: annihilation operator for particles
- $b^{s\dagger}_{\mathbf{p}}$: creation operator for antiparticles
- $s = 1, 2$: spin indices

3.3.1 Anticommutation Relations

For fermions, we impose following **anticommutation relations** to be used for other proofs:

$$\{a^r_{\mathbf{p}}, a^{s\dagger}_{\mathbf{q}}\} = \{b^r_{\mathbf{p}}, b^{s\dagger}_{\mathbf{q}}\} = (2\pi)^3 \delta^3(\mathbf{p} - \mathbf{q}) \delta^{rs} \tag{3.18}$$

$$\{a^r_{\mathbf{p}}, a^s_{\mathbf{q}}\} = \{a^{r\dagger}_{\mathbf{p}}, a^{s\dagger}_{\mathbf{q}}\} = 0$$
$$\{b^r_{\mathbf{p}}, b^s_{\mathbf{q}}\} = \{b^{r\dagger}_{\mathbf{p}}, b^{s\dagger}_{\mathbf{q}}\} = 0$$

All other anticommutators between a and b operators vanish.

3.3.2 Field Anticommutation Relations

$$\boxed{\{\psi(\mathbf{x}, t), \psi^\dagger(\mathbf{y}, t)\} = \delta^3(\mathbf{x} - \mathbf{y})} \tag{3.19}$$

Proof Of Field Anticommutation Relations

We know that the field expansions are :

$$\psi_a(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_p}} \sum_{s=1,2} \left[a^s_{\mathbf{p}} u^s_a(p) e^{-ip\cdot x} + b^{s\dagger}_{\mathbf{p}} v^s_a(p) e^{ip\cdot x} \right]$$

$$\psi^\dagger_b(y) = \int \frac{d^3q}{(2\pi)^3} \frac{1}{\sqrt{2E_q}} \sum_{r=1,2} \left[a^{r\dagger}_{\mathbf{q}} u^{r\dagger}_b(q) e^{iq\cdot y} + b^r_{\mathbf{q}} v^{r\dagger}_b(q) e^{-iq\cdot y} \right]$$

Now compute the anticommutator using these field expansions;

$$\{\psi_a(x), \psi^\dagger_b(y)\} = \int \frac{d^3p\, d^3q}{(2\pi)^6} \frac{1}{\sqrt{4E_pE_q}}$$

$$\times \sum_{r,s} \Big[a^s_{\mathbf{p}} a^{r\dagger}_{\mathbf{q}} u^s_a(p) u^{r\dagger}_b(q) e^{-ip\cdot x + iq\cdot y} + a^{r\dagger}_{\mathbf{q}} a^s_{\mathbf{p}} u^s_a(p) u^{r\dagger}_b(q) e^{-ip\cdot x + iq\cdot y}$$

$$+ a^s_{\mathbf{p}} b^r_{\mathbf{q}} u^s_a(p) v^{r\dagger}_b(q) e^{-ip\cdot x - iq\cdot y} + b^r_{\mathbf{q}} a^s_{\mathbf{p}} u^s_a(p) v^{r\dagger}_b(q) e^{-ip\cdot x - iq\cdot y}$$

$$+ b^{s\dagger}_{\mathbf{p}} a^{r\dagger}_{\mathbf{q}} v^s_a(p) u^{r\dagger}_b(q) e^{ip\cdot x + iq\cdot y} + a^{r\dagger}_{\mathbf{q}} b^{s\dagger}_{\mathbf{p}} v^s_a(p) u^{r\dagger}_b(q) e^{ip\cdot x + iq\cdot y}$$

$$+ b^{s\dagger}_{\mathbf{p}} b^r_{\mathbf{q}} v^s_a(p) v^{r\dagger}_b(q) e^{ip\cdot x - iq\cdot y} + b^r_{\mathbf{q}} b^{s\dagger}_{\mathbf{p}} v^s_a(p) v^{r\dagger}_b(q) e^{ip\cdot x - iq\cdot y} \Big]$$

Now using the anticommutation relations:

$$\{a^s_{\mathbf{p}}, a^{r\dagger}_{\mathbf{q}}\} = (2\pi)^3 \delta^3(\mathbf{p} - \mathbf{q}) \delta^{rs}, \quad \{b^{s\dagger}_{\mathbf{p}}, b^r_{\mathbf{q}}\} = (2\pi)^3 \delta^3(\mathbf{p} - \mathbf{q}) \delta^{rs}$$

$$\{a^s_{\mathbf{p}}, b^r_{\mathbf{q}}\} = 0, \quad \{b^{s\dagger}_{\mathbf{p}}, a^{r\dagger}_{\mathbf{q}}\} = 0$$

The non-vanishing terms become:

$$\{\psi_a(x), \psi^\dagger_b(y)\} = \int \frac{d^3p\, d^3q}{(2\pi)^6} \frac{1}{\sqrt{4E_pE_q}}$$

$$\times \sum_{r,s} \Big[(2\pi)^3 \delta^3(\mathbf{p} - \mathbf{q}) \delta^{rs} u^s_a(p) u^{r\dagger}_b(q) e^{-ip\cdot x + iq\cdot y} + (2\pi)^3 \delta^3(\mathbf{p} - \mathbf{q}) \delta^{rs} v^s_a(p) v^{r\dagger}_b(q) e^{ip\cdot x - iq\cdot y} \Big]$$

Integrating over $\mathbf{q}$:

$$\{\psi_a(x), \psi^\dagger_b(y)\} = \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_p} \sum_s \left[u^s_a(p) u^{s\dagger}_b(p) e^{-ip\cdot(x-y)} + v^s_a(p) v^{s\dagger}_b(p) e^{ip\cdot(x-y)} \right]$$

Using the completeness relations from (3.14) and (3.15), we have:

$$\sum_s u^s_a(p) u^{s\dagger}_b(p) = [(\not{p} + m)\gamma^0]_{ab}, \quad \sum_s v^s_a(p) v^{s\dagger}_b(p) = [(\not{p} - m)\gamma^0]_{ab}$$

Substituting:

$$\{\psi_a(x), \psi_b^\dagger(y)\} = \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_p} \left[[(\not{p} + m)\gamma^0]_{ab} e^{-ip\cdot(x-y)} + [(\not{p} - m)\gamma^0]_{ab} e^{ip\cdot(x-y)} \right]$$

At equal times $x^0 = y^0$, we have:

$$\{\psi_a(\mathbf{x}, t), \psi_b^\dagger(\mathbf{y}, t)\} = \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_p} \left[[(\not{p} + m)\gamma^0]_{ab} e^{-i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} + [(\not{p} - m)\gamma^0]_{ab} e^{i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} \right]$$

now writing: $\not{p} = \gamma^0 E_p - \vec{\gamma} \cdot \mathbf{p}$:
First term:

$$[(\not{p} + m)\gamma^0]_{ab} = [(\gamma^0 E_p - \vec{\gamma} \cdot \mathbf{p} + m)\gamma^0]_{ab} = [E_p \gamma^0 \gamma^0 - (\vec{\gamma} \cdot \mathbf{p})\gamma^0 + m\gamma^0]_{ab}$$

$$= [E_p I - (\vec{\gamma} \cdot \mathbf{p})\gamma^0 + m\gamma^0]_{ab}$$

Second term:

$$[(\not{p} - m)\gamma^0]_{ab} = [(\gamma^0 E_p - \vec{\gamma} \cdot \mathbf{p} - m)\gamma^0]_{ab} = [E_p I - (\vec{\gamma} \cdot \mathbf{p})\gamma^0 - m\gamma^0]_{ab}$$

Now the full expression becomes:

$$\{\psi_a, \psi_b^\dagger\} = \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_p} \left[[E_p I - (\vec{\gamma} \cdot \mathbf{p})\gamma^0 + m\gamma^0]_{ab} e^{-i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} + [E_p I - (\vec{\gamma} \cdot \mathbf{p})\gamma^0 - m\gamma^0]_{ab} e^{i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} \right]$$

Making the substitution $\mathbf{p} \to -\mathbf{p}$ in the first term:

$$\{\psi_a, \psi_b^\dagger\} = \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_p} \left[[E_p I + (\vec{\gamma} \cdot \mathbf{p})\gamma^0 + m\gamma^0]_{ab} e^{i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} + [E_p I - (\vec{\gamma} \cdot \mathbf{p})\gamma^0 - m\gamma^0]_{ab} e^{i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} \right]$$

Therefore,

$$\{\psi_a, \psi_b^\dagger\} = \int \frac{d^3p}{(2\pi)^3} \delta_{ab} e^{i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} = \delta^3(\mathbf{x} - \mathbf{y})\delta_{ab}$$

3.3.3 Quantized Dirac Hamiltonian

We are now prepared to formulate the quantum theory corresponding to the free Dirac field, starting from its Lagrangian density given by:

$$\mathcal{L} = \bar{\psi}(i\gamma^\mu \partial_\mu - m)\psi,$$

the canonical momentum conjugate to ψ is $\pi = \frac{\partial \mathcal{L}}{\partial \dot{\psi}} = i\psi^\dagger$, and thus the total Hamiltonian is

$$H = \int d^3x\,(\pi\dot{\psi} - \mathcal{L}) = \int d^3x\,\bar{\psi}(-i\vec{\gamma}\cdot\nabla + m)\psi = \int d^3x\,\psi^\dagger[-i\gamma^0\vec{\gamma}\cdot\nabla + m\gamma^0]\psi.$$

Since $\vec{\alpha} = \gamma^0\vec{\gamma}$, $\beta = \gamma^0$, the quantity in brackets is the Dirac Hamiltonian ;

$$h_D = -i\vec{\alpha}\cdot\nabla + m\beta. \tag{3.20}$$

Using the field expansion:

$$\psi(x) = \int \frac{d^3p}{(2\pi)^3}\frac{1}{\sqrt{2E_p}}\sum_{s=1,2}\left[a^s_{\mathbf{p}}u^s(p)e^{-ip\cdot x} + b^{s\dagger}_{\mathbf{p}}v^s(p)e^{ip\cdot x}\right]$$

Now we evaluate $h_D\psi(x)$:
First we evaluate the spatial derivative:

$$\nabla\psi(x) = \int \frac{d^3p}{(2\pi)^3}\frac{1}{\sqrt{2E_p}}\sum_{s=1,2}\left[a^s_{\mathbf{p}}(i\mathbf{p})u^s(p)e^{-ip\cdot x} + b^{s\dagger}_{\mathbf{p}}(-i\mathbf{p})v^s(p)e^{ip\cdot x}\right]$$

Then:

$$h_D\psi(x) = \int \frac{d^3p}{(2\pi)^3}\frac{1}{\sqrt{2E_p}}\sum_{s=1,2}\left[a^s_{\mathbf{p}}(\vec{\alpha}\cdot\mathbf{p} + m\beta)u^s(p)e^{-ip\cdot x} + b^{s\dagger}_{\mathbf{p}}(-\vec{\alpha}\cdot\mathbf{p} + m\beta)v^s(p)e^{ip\cdot x}\right]$$

Using the Dirac equation:
For positive energy solutions $u^s(p)$:

$$(i\gamma^\mu\partial_\mu - m)u^s(p)e^{-ip\cdot x} = 0 \Rightarrow (\gamma^\mu p_\mu - m)u^s(p) = 0$$

$$\Rightarrow (\gamma^0 E_p - \vec{\gamma}\cdot\mathbf{p} - m)u^s(p) = 0$$

Multiply by γ^0 from the left:

$$\gamma^0(\gamma^0 E_p - \vec{\gamma}\cdot\mathbf{p} - m)u^s(p) = (E_p - \gamma^0\vec{\gamma}\cdot\mathbf{p} - m\gamma^0)u^s(p) = 0$$

Using $\vec{\alpha} = \gamma^0\vec{\gamma}$ and $\beta = \gamma^0$:

$$(E_p - \vec{\alpha}\cdot\mathbf{p} - m\beta)u^s(p) = 0 \Rightarrow (\vec{\alpha}\cdot\mathbf{p} + m\beta)u^s(p) = E_p u^s(p)$$

For negative energy solutions $v^s(p)$:

$$(i\gamma^\mu\partial_\mu - m)v^s(p)e^{ip\cdot x} = 0 \Rightarrow (\gamma^\mu(-p_\mu) - m)v^s(p) = 0$$

$$\Rightarrow (-\gamma^0 E_p - \vec{\gamma}\cdot\mathbf{p} - m)v^s(p) = 0$$

Multiply by γ^0 from the left:

$$\gamma^0(-\gamma^0 E_p - \vec{\gamma}\cdot\mathbf{p} - m)v^s(p) = (-E_p - \gamma^0\vec{\gamma}\cdot\mathbf{p} - m\gamma^0)v^s(p) = 0$$

Using $\vec{\alpha} = \gamma^0\vec{\gamma}$ and $\beta = \gamma^0$:

$$(-E_p - \vec{\alpha}\cdot\mathbf{p} - m\beta)v^s(p) = 0 \Rightarrow (-\vec{\alpha}\cdot\mathbf{p} + m\beta)v^s(p) = -E_p v^s(p)$$

Therefore:

$$h_D\psi(x) = \int \frac{d^3p}{(2\pi)^3}\frac{1}{\sqrt{2E_p}}\sum_{s=1,2}\left[a^s_{\mathbf{p}}E_p u^s(p)e^{-ip\cdot x} - b^{s\dagger}_{\mathbf{p}}E_p v^s(p)e^{ip\cdot x}\right]$$

Now since the Hamiltonian is

$$H = \int d^3x\, \psi^\dagger h_D\psi$$

Substitute both $\psi^\dagger$ and $h_D\psi$ in H;

$$\psi^\dagger(x) = \int \frac{d^3q}{(2\pi)^3}\frac{1}{\sqrt{2E_q}}\sum_{r=1,2}\left[a^{r\dagger}_{\mathbf{q}}u^{r\dagger}(q)e^{iq\cdot x} + b^r_{\mathbf{q}}v^{r\dagger}(q)e^{-iq\cdot x}\right]$$

$$H = \int d^3x \int \frac{d^3p d^3q}{(2\pi)^6}\frac{1}{\sqrt{4E_pE_q}}\sum_{r,s}E_p$$

$$\times\left[a^{r\dagger}_{\mathbf{q}}a^s_{\mathbf{p}}u^{r\dagger}(q)u^s(p)e^{i(q-p)\cdot x} - a^{r\dagger}_{\mathbf{q}}b^{s\dagger}_{\mathbf{p}}u^{r\dagger}(q)v^s(p)e^{i(q+p)\cdot x}\right.$$

$$\left.+b^r_{\mathbf{q}}a^s_{\mathbf{p}}v^{r\dagger}(q)u^s(p)e^{-i(q+p)\cdot x} - b^r_{\mathbf{q}}b^{s\dagger}_{\mathbf{p}}v^{r\dagger}(q)v^s(p)e^{-i(q-p)\cdot x}\right]$$

Now integrating over d^3x using; we get:

$$\int d^3x\, e^{i(\mathbf{q}\pm\mathbf{p})\cdot\mathbf{x}} = (2\pi)^3\delta^3(\mathbf{q}\pm\mathbf{p})$$

$$H = \int \frac{d^3p}{(2\pi)^3}\frac{1}{2E_p}\sum_{r,s}E_p\left[a^{r\dagger}_{\mathbf{p}}a^s_{\mathbf{p}}u^{r\dagger}(p)u^s(p) - b^r_{\mathbf{p}}b^{s\dagger}_{\mathbf{p}}v^{r\dagger}(p)v^s(p)\right]$$

Using the orthogonality relations ; we have now

$$u^{r\dagger}(p)u^s(p) = 2E_p\delta^{rs}, \quad v^{r\dagger}(p)v^s(p) = 2E_p\delta^{rs}$$

So we obtain obtain;

$$H = \int \frac{d^3p}{(2\pi)^3} E_p \sum_{s=1,2} \left[a_{\mathbf{p}}^{s\dagger} a_{\mathbf{p}}^{s} - b_{\mathbf{p}}^{s} b_{\mathbf{p}}^{s\dagger} \right]$$

Using the anticommutation relation $\{b_{\mathbf{p}}^{s}, b_{\mathbf{p}}^{s\dagger}\} = (2\pi)^3 \delta^3(0)$;

$$H = \int \frac{d^3p}{(2\pi)^3} E_p \sum_{s=1,2} \left[a_{\mathbf{p}}^{s\dagger} a_{\mathbf{p}}^{s} + b_{\mathbf{p}}^{s\dagger} b_{\mathbf{p}}^{s} \right] - 2E_p \delta^3(0)$$

The last term is the infinite zero-point energy. If we use normal ordering, then

$$\boxed{: H := \int \frac{d^3p}{(2\pi)^3} E_{\mathbf{p}} \sum_{s=1,2} \left[a_{\mathbf{p}}^{s\dagger} a_{\mathbf{p}}^{s} + b_{\mathbf{p}}^{s\dagger} b_{\mathbf{p}}^{s} \right]} \tag{3.21}$$

For the normal ordered Hamiltonian:

$$\langle 0| : H : |0\rangle = 0$$

This shows that the vacuum energy is zero after normal ordering, eliminating the infinite zero-point energy.

This is the **quantized Dirac Hamiltonian**.

3.4 Conserved Current for the Dirac Field

The free Dirac Lagrangian density is given by

$$\mathcal{L} = \bar{\psi}(i\gamma^\mu \partial_\mu - m)\psi = i\bar{\psi}\gamma^\mu \partial_\mu \psi - m\bar{\psi}\psi. \tag{3.22}$$

Global $U(1)$ Symmetry

The Lagrangian is invariant under the global $U(1)$ phase transformation

$$\begin{aligned} \psi &\to e^{iq}\psi, & \Rightarrow \qquad \delta\psi &= iq\psi, \\ \bar{\psi} &\to e^{-iq}\bar{\psi}, & \Rightarrow \qquad \delta\bar{\psi} &= -iq\bar{\psi}, \end{aligned}$$

where q is a real constant parameter representing the charge.

Noether Current

According to Noether's theorem, the conserved current associated with this continuous symmetry is

$$j^\mu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \psi)}\,\delta\psi + \delta\bar{\psi}\,\frac{\partial \mathcal{L}}{\partial(\partial_\mu \bar{\psi})}.$$

From the Lagrangian, we have

$$\frac{\partial \mathcal{L}}{\partial(\partial_\mu \psi)} = i\bar{\psi}\gamma^\mu, \qquad \frac{\partial \mathcal{L}}{\partial(\partial_\mu \bar{\psi})} = 0.$$

Substituting these and the variations of the fields:

$$\begin{aligned} j^\mu &= (i\bar{\psi}\gamma^\mu)(iq\psi) + (-iq\bar{\psi})(0), \\ &= -q\,\bar{\psi}\gamma^\mu\psi. \end{aligned}$$

The overall sign is conventional; we therefore define the conserved current as

$$\boxed{j^\mu = q\,\bar{\psi}\gamma^\mu\psi.} \tag{3.23}$$

Conservation Law

The conservation of the current can be shown using the Dirac equation and its adjoint:

$$\begin{aligned} (i\gamma^\mu\partial_\mu - m)\psi &= 0, \\ \bar{\psi}(i\overleftarrow{\partial}_\mu\gamma^\mu + m) &= 0. \end{aligned}$$

Differentiating the current,

$$\partial_\mu j^\mu = q\left[(\partial_\mu\bar{\psi})\gamma^\mu\psi + \bar{\psi}\gamma^\mu\partial_\mu\psi\right],$$

and substituting the equations of motion gives

$$\partial_\mu j^\mu = q\left[-im\bar{\psi}\psi + im\bar{\psi}\psi\right] = 0.$$

Hence, the current is conserved:

$$\boxed{\partial_\mu j^\mu = 0.} \tag{3.24}$$

Conserved Charge

The corresponding conserved charge is

$$Q = \int d^3x\, j^0 = q \int d^3x\, \bar{\psi}\gamma^0\psi = q \int d^3x\, \psi^\dagger\psi, \tag{3.25}$$

which represents the total charge (or fermion number) associated with the Dirac field.

3.4.1 Quantized Form Of Conserved Charge

Using the field decompositions for ψ and $\psi^\dagger$:

$$\psi(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_p}} \sum_{s=1,2} \left[a^s_{\mathbf{p}} u^s(p) e^{-ip\cdot x} + b^{s\dagger}_{\mathbf{p}} v^s(p) e^{ip\cdot x} \right]$$

$$\psi^\dagger(x) = \int \frac{d^3q}{(2\pi)^3} \frac{1}{\sqrt{2E_q}} \sum_{r=1,2} \left[a^{r\dagger}_{\mathbf{q}} u^{r\dagger}(q) e^{iq\cdot x} + b^r_{\mathbf{q}} v^{r\dagger}(q) e^{-iq\cdot x} \right]$$

From equation (3.25), the conserved charge operator is:

$$Q = q \int d^3x\, \psi^\dagger(x)\psi(x)$$

Substituting the field expansions;

$$Q = q \int d^3x \int \frac{d^3p d^3q}{(2\pi)^6} \frac{1}{\sqrt{4E_pE_q}} \sum_{r,s}$$
$$\times \Big[a^{r\dagger}_{\mathbf{q}} a^s_{\mathbf{p}} u^{r\dagger}(q) u^s(p) e^{i(q-p)\cdot x} + a^{r\dagger}_{\mathbf{q}} b^{s\dagger}_{\mathbf{p}} u^{r\dagger}(q) v^s(p) e^{i(q+p)\cdot x}$$
$$+ b^r_{\mathbf{q}} a^s_{\mathbf{p}} v^{r\dagger}(q) u^s(p) e^{-i(q+p)\cdot x} + b^r_{\mathbf{q}} b^{s\dagger}_{\mathbf{p}} v^{r\dagger}(q) v^s(p) e^{-i(q-p)\cdot x} \Big]$$

Integrate over d^3x using:

$$\int d^3x\, e^{i(\mathbf{q}\pm\mathbf{p})\cdot\mathbf{x}} = (2\pi)^3 \delta^3(\mathbf{q}\pm\mathbf{p})$$

We get:

$$Q = q \int \frac{d^3p d^3q}{(2\pi)^6} \frac{1}{\sqrt{4E_pE_q}} \sum_{r,s} (2\pi)^3$$
$$\times \Big[a^{r\dagger}_{\mathbf{q}} a^s_{\mathbf{p}} u^{r\dagger}(q) u^s(p) \delta^3(\mathbf{q}-\mathbf{p}) + a^{r\dagger}_{\mathbf{q}} b^{s\dagger}_{\mathbf{p}} u^{r\dagger}(q) v^s(p) \delta^3(\mathbf{q}+\mathbf{p})$$
$$+ b^r_{\mathbf{q}} a^s_{\mathbf{p}} v^{r\dagger}(q) u^s(p) \delta^3(\mathbf{q}+\mathbf{p}) + b^r_{\mathbf{q}} b^{s\dagger}_{\mathbf{p}} v^{r\dagger}(q) v^s(p) \delta^3(\mathbf{q}-\mathbf{p}) \Big]$$

Now integrating over $\mathbf{q}$ using the delta functions:
$\delta^3(\mathbf{q}-\mathbf{p})$: $\mathbf{q}=\mathbf{p}$, so $E_q = E_p$ and $\delta^3(\mathbf{q}+\mathbf{p})$: $\mathbf{q}=-\mathbf{p}$, so $E_q = E_p$

Thus:
The 2nd and 3rd terms vanish due to the orthogonality of Dirac spinors:
Using the spinor orthogonality relation:

$$u^{r\dagger}(-\mathbf{p})v^s(p) = 0$$

and

$$v^{r\dagger}(-\mathbf{p})u^s(p) = 0$$

so we get;

$$Q = q\int \frac{d^3p}{(2\pi)^3}\frac{1}{2E_p}\sum_{r,s}\left[a^{r\dagger}_{\mathbf{p}}a^s_{\mathbf{p}}u^{r\dagger}(p)u^s(p) + b^r_{\mathbf{p}}b^{s\dagger}_{\mathbf{p}}v^{r\dagger}(p)v^s(p)\right]$$

Using the orthogonality relations:

$$u^{r\dagger}(p)u^s(p) = 2E_p\delta^{rs}, \quad v^{r\dagger}(p)v^s(p) = 2E_p\delta^{rs}$$

We obtain:

$$Q = q\int \frac{d^3p}{(2\pi)^3}\sum_{s=1,2}\left[a^{s\dagger}_{\mathbf{p}}a^s_{\mathbf{p}} + b^s_{\mathbf{p}}b^{s\dagger}_{\mathbf{p}}\right]$$

Using the anticommutation relation $\{b^s_{\mathbf{p}}, b^{s\dagger}_{\mathbf{p}}\} = (2\pi)^3\delta^3(0)$; we get :

$$Q = q\int \frac{d^3p}{(2\pi)^3}\sum_{s=1,2}\left[a^{s\dagger}_{\mathbf{p}}a^s_{\mathbf{p}} - b^{s\dagger}_{\mathbf{p}}b^s_{\mathbf{p}}\right] + q\delta^3(0)$$

The last term is an infinite constant. Using normal ordering:

$$\boxed{:Q:= q\int \frac{d^3p}{(2\pi)^3}\sum_{s=1,2}\left[a^{s\dagger}_{\mathbf{p}}a^s_{\mathbf{p}} - b^{s\dagger}_{\mathbf{p}}b^s_{\mathbf{p}}\right]} \tag{3.26}$$

This represents the total charge operator, where $a^{s\dagger}_{\mathbf{p}}a^s_{\mathbf{p}}$ counts particles and $b^{s\dagger}_{\mathbf{p}}b^s_{\mathbf{p}}$ counts antiparticles with opposite charge, they both have same charge of magnitude q with opposite sign.

3.5 Dirac Propagator

We know that the field $\psi_a(x)$ and $\bar{\psi}_b(y)$ are given by:

$$\psi_a(x) = \int \frac{d^3p}{(2\pi)^3}\frac{1}{\sqrt{2E_p}}\sum_{s=1,2}\left[a^s_{\mathbf{p}}u^s_a(p)e^{-ip\cdot x} + b^{s\dagger}_{\mathbf{p}}v^s_a(p)e^{ip\cdot x}\right]$$

$$\bar{\psi}_b(y) = \int \frac{d^3q}{(2\pi)^3} \frac{1}{\sqrt{2E_q}} \sum_{r=1,2} \left[a^{r\dagger}_{\mathbf{q}} \bar{u}^r_b(q) e^{iq\cdot y} + b^r_{\mathbf{q}} \bar{v}^r_b(q) e^{-iq\cdot y} \right]$$

We now evaluate the vacuum expectation value(VEV) of $\psi_a(x)\bar{\psi}_b(y)$ as follows:

$$\langle 0|\psi_a(x)\bar{\psi}_b(y)|0\rangle = \int \frac{d^3p d^3q}{(2\pi)^6} \frac{1}{\sqrt{4E_pE_q}} \sum_{r,s}$$

$$\times \Big[\langle 0|a^s_{\mathbf{p}} a^{r\dagger}_{\mathbf{q}}|0\rangle u^s_a(p)\bar{u}^r_b(q) e^{-ip\cdot x+iq\cdot y} + \langle 0|a^s_{\mathbf{p}} b^r_{\mathbf{q}}|0\rangle u^s_a(p)\bar{v}^r_b(q) e^{-ip\cdot x-iq\cdot y}$$

$$+ \langle 0|b^{s\dagger}_{\mathbf{p}} a^{r\dagger}_{\mathbf{q}}|0\rangle v^s_a(p)\bar{u}^r_b(q) e^{ip\cdot x+iq\cdot y} + \langle 0|b^{s\dagger}_{\mathbf{p}} b^r_{\mathbf{q}}|0\rangle v^s_a(p)\bar{v}^r_b(q) e^{ip\cdot x-iq\cdot y} \Big]$$

We know that

$$\langle 0|a^s_{\mathbf{p}} a^{r\dagger}_{\mathbf{q}}|0\rangle = (2\pi)^3 \delta^{rs} \delta^3(\mathbf{p}-\mathbf{q})$$

$$\langle 0|a^s_{\mathbf{p}} b^r_{\mathbf{q}}|0\rangle = 0$$

$$\langle 0|b^{s\dagger}_{\mathbf{p}} a^{r\dagger}_{\mathbf{q}}|0\rangle = 0$$

$$\langle 0|b^{s\dagger}_{\mathbf{p}} b^r_{\mathbf{q}}|0\rangle = 0$$

using these results ,we have only the first term surviving:

$$\langle 0|\psi_a(x)\bar{\psi}_b(y)|0\rangle = \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_p} \sum_s u^s_a(p)\bar{u}^s_b(p) e^{-ip\cdot(x-y)}$$

Using the completeness relation:

$$\sum_s u^s_a(p)\bar{u}^s_b(p) = (\not{p} + m)_{ab}$$

We obtain:

$$\langle 0|\psi_a(x)\bar{\psi}_b(y)|0\rangle = \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_p} (\not{p} + m)_{ab} e^{-ip\cdot(x-y)}$$

$$= (i\not{\partial}_x + m)_{ab} \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_p} e^{-ip\cdot(x-y)} \tag{3.27}$$

Now we calculate the vacuum expectation value of $\bar{\psi}_b(y)\psi_a(x)$:

$$\langle 0|\bar{\psi}_b(y)\psi_a(x)|0\rangle = \int \frac{d^3p d^3q}{(2\pi)^6} \frac{1}{\sqrt{4E_pE_q}} \sum_{r,s}$$

$$\times \Big[\langle 0|a^{r\dagger}_{\mathbf{q}} a^s_{\mathbf{p}}|0\rangle \bar{u}^r_b(q) u^s_a(p) e^{iq\cdot y-ip\cdot x} + \langle 0|a^{r\dagger}_{\mathbf{q}} b^{s\dagger}_{\mathbf{p}}|0\rangle \bar{u}^r_b(q) v^s_a(p) e^{iq\cdot y+ip\cdot x}$$

$$+ \langle 0|b^r_{\mathbf{q}} a^s_{\mathbf{p}}|0\rangle \bar{v}^r_b(q) u^s_a(p) e^{-iq\cdot y-ip\cdot x} + \langle 0|b^r_{\mathbf{q}} b^{s\dagger}_{\mathbf{p}}|0\rangle \bar{v}^r_b(q) v^s_a(p) e^{-iq\cdot y+ip\cdot x} \Big]$$

using the following results;

$$\langle 0|a^{r\dagger}_{\mathbf{q}} a^{s}_{\mathbf{p}}|0\rangle = 0$$

$$\langle 0|a^{r\dagger}_{\mathbf{q}} b^{s\dagger}_{\mathbf{p}}|0\rangle = 0$$

$$\langle 0|b^{r}_{\mathbf{q}} a^{s}_{\mathbf{p}}|0\rangle = 0$$

$$\langle 0|b^{r}_{\mathbf{q}} b^{s\dagger}_{\mathbf{p}}|0\rangle = \langle 0|[b^{r}_{\mathbf{q}}, b^{s\dagger}_{\mathbf{p}}]|0\rangle = (2\pi)^3 \delta^{rs} \delta^3(\mathbf{p}-\mathbf{q})$$

we get ;

$$\langle 0|\bar{\psi}_b(y)\psi_a(x)|0\rangle = \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_p} \sum_s \bar{v}^s_b(p) v^s_a(p) e^{ip\cdot(x-y)}$$

Using the completeness relation:

$$\sum_s v^s_a(p)\bar{v}^s_b(p) = (\not{p} - m)_{ab}$$

so we obtain:

$$\langle 0|\bar{\psi}_b(y)\psi_a(x)|0\rangle = -(i\not{\partial}_x - m)_{ab} \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_p} e^{ip\cdot(x-y)} \tag{3.28}$$

By combining the two propagators (3.27) and (3.28), we get the **Feynman Propagator** as follow:

$$S_F(x-y) = \langle 0|T[\psi(x)\bar{\psi}(y)]|0\rangle$$

where T is the **time-ordering operator** for fermionic fields:

$$T[\psi(x)\bar{\psi}(y)] = \begin{cases} \psi(x)\bar{\psi}(y) & \text{for } x^0 > y^0 \\ -\bar{\psi}(y)\psi(x) & \text{for } x^0 < y^0 \end{cases}$$

Therefore, the Feynman propagator for the Dirac field is:

$$S_F(x-y) = \begin{cases} \langle 0|\psi(x)\bar{\psi}(y)|0\rangle & \text{for } x^0 > y^0 \\ -\langle 0|\bar{\psi}(y)\psi(x)|0\rangle & \text{for } x^0 < y^0 \end{cases} \tag{3.29}$$

Using the equations (3.27) and (3.28), we can write the Dirac Propagator as :

$$S_F(x-y) = \begin{cases} (i\not{\partial}_x + m)_{ab} \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_p} e^{-ip\cdot(x-y)} & \text{for } x^0 > y^0 \\ -(i\not{\partial}_x - m)_{ab} \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_p} e^{ip\cdot(x-y)} & \text{for } x^0 < y^0 \end{cases} \tag{3.30}$$

By taking Fourier Transform of Dirac propagator for the case of $x^0 > y^0$, we have:

$$S_F(x-y) = \int \frac{d^4p}{(2\pi)^4} e^{-ip\cdot(x-y)} \widetilde{S_F}(p)$$

Now operate by $(i\partial\!\!\!/_x - m)$ on both sides:
Left side:

$$(i\partial\!\!\!/_x - m) S_F(x-y) = i\delta^4(x-y)$$

Right side:

$$(i\partial\!\!\!/_x - m) \int \frac{d^4p}{(2\pi)^4} e^{-ip\cdot(x-y)} \widetilde{S_F}(p) = (i(-ip\!\!\!/) - m) \int \frac{d^4p}{(2\pi)^4} e^{-ip\cdot(x-y)} \widetilde{S_F}(p)$$

We know that the Fourier representation of the delta function is:

$$\delta^4(x-y) = \int \frac{d^4p}{(2\pi)^4} e^{-ip\cdot(x-y)}$$

Comparing both sides, we identify:

$$(p\!\!\!/ - m)\widetilde{S_F}(p) = i$$

So we get:

$$\widetilde{S_F}(p) = \frac{i}{(p\!\!\!/ - m)}$$

multiplying numerator and denominator by $(p\!\!\!/ + m), weget;$

$$\widetilde{S_F}(p) = \frac{i(p\!\!\!/ + m)}{(p^2 - m^2)}$$

using Feynman $i\epsilon$ prescription,we can write

$$\boxed{S_F(x-y) = \int \frac{d^4p}{(2\pi)^4} \frac{i(p\!\!\!/ + m)}{p^2 - m^2 + i\epsilon} e^{-ip\cdot(x-y)}} \tag{3.31}$$

This satisfies $(\mathrm{i}\partial_x - m)S_F(x-y) = i\delta^{(4)}(x-y)$, so that S_F is a Green's function for the Dirac operator.

Chapter–3 Summary

In this chapter,we derived the Dirac equation, and constructed the quantum field operator via Fourier mode expansion. The Feynman propagator $S_F(x-y)$ was obtained, and we derived the Hamiltonian : $H := \int d^3p\, E_p \sum_s \left[a^{s\dagger}_{\mathbf{p}} a^s_{\mathbf{p}} + b^{s\dagger}_{\mathbf{p}} b^s_{\mathbf{p}}\right]$. This formalism establishes the Dirac field as a theory of spin-1/2 particles and antiparticles.

Chapter 4

Fields Interactions And Feynman Diagrams

4.1 Introduction

To describe the real world, interaction terms must be added to the free Hamiltonian or Lagrangian i

$$H_{\text{int}} = \int d^3x \mathcal{H}_{\text{int}}[\phi(x)] = -\int d^3x \mathcal{L}_{\text{int}}[\phi(x)]$$

where $\mathcal{H}_{\text{int}} = -\mathcal{L}_{\text{int}}$ is a function only of fields, not their derivatives.

(so the corresponding conjugate momenta are zero).

To preserve causality, only the products of different fields at the same spacetime points are allow

therefore, $[\phi(x)]^4$ is allowed but $\phi(x)\phi(y)$ is not allowed for the interaction Lagrangian.

4.1.1 Canonical Dimensions

In the natural units, we take:

$$\hbar = c = 1.$$

Here, $\hbar$ has units of **J.s** and c has units of **m/s**. Therefore,by dimensional analysis ;

$$[L] = [T] = [M]^{-1}$$

So ,in these units, mass, energy, and momentum have the same dimensions, while length and time carry inverse mass dimension.

Examples

1. The **action**

$$S = \int d^4x \, \mathcal{L}$$

is dimensionless in the natural units because Lagrangian has dimension of energy

and so ;

$$[S] = [d^4x]^4[\mathcal{L}]^4 = [L]^{-4}[E]^4 = [M]^{-4}[M]^4 = [M]^0$$

2.The interaction part Lagrangian of ϕ^4-theory is

$$\mathcal{L}_{\text{int}} = -\frac{\lambda}{4!}\phi^4 \tag{4.2}$$

Since the field ϕ has dimension of $[M]$ and so

$$[M]^4 = [\lambda][M]^4$$

$$[\lambda] = [M]^0$$

(4.3)

therefore, the coupling constant λ is dimensionless.

4.2 Perturbation expansion and Wick's Theorem

In this section, i have explained the Dyson series and The Wick's theorem in detailed mathematical derivations and their usefulness in Feynman diagrams.

4.2.1 Interaction Picture Fields

At any fixed time t, the scalar field in the interaction picture is given by:

$$\phi(t,\vec{x}) = \int \frac{d^3p}{(2\pi)^3}\frac{1}{\sqrt{2E_p}}\left[a_p e^{-ip\cdot x} + a_p^\dagger e^{ip\cdot x}\right] = e^{iH(t-t_0)}\phi(t_0,\vec{x})e^{-iH(t-t_0)} \tag{4.4}$$

The Hamiltonian in presence of interaction is given by

$$H = H_0 + H_{\text{int}}$$

If $\lambda = 0$, then $H = H_0$ and so now:

$$\phi(t,\vec{x}) = e^{iH_0(t-t_0)}\phi(t_0,\vec{x})e^{-iH_0(t-t_0)} = \phi_I(t,\vec{x}) \tag{4.5}$$

where $\phi_I(t,\vec{x})$ is called the **interaction-picture field** .
Now we can write equation (4.5) in terms of unitary operators as:

$$\phi(t,\vec{x}) = e^{iH(t-t_0)}e^{-iH_0(t-t_0)}\phi_I(t,\vec{x})e^{iH_0(t-t_0)}e^{-iH(t-t_0)}$$

$= \mathrm{U}^{\dagger}(t,t_0)\phi_I(t,\vec{x})U(t,t_0)$ where

$$U(t,t_0) = e^{iH_0(t-t_0)}e^{-iH(t-t_0)} \tag{4.7}$$

and

$$U(t_0,t_0) = 1 \tag{4.8}$$

Now, by differentiating equation (4.7), we get:

$$\begin{aligned}
i\frac{\partial}{\partial t}U(t,t_0) &= i\frac{\partial}{\partial t}\left[e^{iH_0(t-t_0)}e^{-iH(t-t_0)}\right] \\
&= i\left[iH_0e^{iH_0(t-t_0)}e^{-iH(t-t_0)} + e^{iH_0(t-t_0)}(-iH)e^{-iH(t-t_0)}\right] \\
&= -H_0e^{iH_0(t-t_0)}e^{-iH(t-t_0)} + He^{iH_0(t-t_0)}e^{-iH(t-t_0)} \\
&= (H-H_0)e^{iH_0(t-t_0)}e^{-iH(t-t_0)} \\
&= H_{\text{int}}e^{iH_0(t-t_0)}e^{-iH(t-t_0)} \\
&= e^{iH_0(t-t_0)}H_{\text{int}}e^{-iH_0(t-t_0)}e^{iH_0(t-t_0)}e^{-iH(t-t_0)} \\
&= e^{iH_0(t-t_0)}H_{\text{int}}e^{-iH_0(t-t_0)}U(t,t_0)
\end{aligned}$$

Therefore, the Schrödinger equation for $U(t,t_0)$ is:

$$\boxed{i\frac{\partial}{\partial t}U(t,t_0) = H_I(t)U(t,t_0)} \tag{4.9}$$

where $H_I(t) = e^{iH_0(t-t_0)}H_{\text{int}}e^{-iH_0(t-t_0)}$ is the interaction Hamiltonian in the interaction picture.

4.2.2 Dyson Series

From equation (4.9), we have

$$i\frac{\partial}{\partial t}U(t,t_0) = H_I(t)U(t,t_0)$$

We can integrate it to get:

$$\int_{t_0}^{t} dU(t',t_0) = -i\int_{t_0}^{t} H_I(t')U(t',t_0)dt'$$

$$U(t,t_0) - U(t_0,t_0) = -i\int_{t_0}^{t} H_I(t')U(t',t_0)dt'$$

Since $U(t_0,t_0) = 1$, we have;

$$U(t,t_0) = 1 - i\int_{t_0}^{t} H_I(t')U(t',t_0)dt' \tag{4.10}$$

Now we put expression of $U(t',t_0)$ in equation (4.10),then

$$U(t,t_0) = 1 - i\int_{t_0}^{t} H_I(t_1)\left[1 - i\int_{t_0}^{t_1} H_I(t_2)U(t_2,t_0)dt_2\right]dt_1$$

or

$$U(t,t_0) = 1 - i\int_{t_0}^{t} H_I(t_1)dt_1 + (-i)^2\int_{t_0}^{t} dt_1\int_{t_0}^{t_1} dt_2\, H_I(t_1)H_I(t_2)U(t_2,t_0)$$

By continuing this iteration indefinitely we get the following **Dyson series**:

$$\begin{aligned} U(t,t_0) &= 1 + \sum_{n=1}^{\infty}(-i)^n\int_{t_0}^{t} dt_1\int_{t_0}^{t_1} dt_2\cdots\int_{t_0}^{t_{n-1}} dt_n\, H_I(t_1)H_I(t_2)\cdots H_I(t_n) \\ &= \sum_{n=0}^{\infty}\frac{(-i)^n}{n!}\int_{t_0}^{t} dt_1\int_{t_0}^{t} dt_2\cdots\int_{t_0}^{t} dt_n\, T\left[H_I(t_1)H_I(t_2)\cdots H_I(t_n)\right] \end{aligned} \tag{4.11}$$

where T is the time-ordering operator,this can be written more compactly as :

$$\boxed{U(t,t_0) = T\exp\left[-i\int_{t_0}^{t} H_I(t')dt'\right]} \tag{4.12}$$

4.2.3 Wick's Theorem

We can write field as the sum of positive and negative frequency parts as :

$$\phi(x) = \phi^+(x) + \phi^-(x) \tag{4.13}$$

where

$$\phi^+(x) = \int\frac{d^3p}{(2\pi)^3}\frac{1}{\sqrt{2E_p}}a_{\vec{p}e^{-ip\cdot x}} \tag{4.14}$$

and

$$\phi^-(x) = \int\frac{d^3p}{(2\pi)^3}\frac{1}{\sqrt{2E_p}}a_{\vec{p}^\dagger e^{ip\cdot x}} \tag{4.15}$$

Now, for $x^0 > y^0$:

$$\phi(x)\phi(y) = [\phi^+(x) + \phi^-(x)][\phi^+(y) + \phi^-(y)]$$

$$= \phi^+(x)\phi^+(y) + \phi^+(x)\phi^-(y) + \phi^-(x)\phi^+(y) + \phi^-(x)\phi^-(y)$$

$$= \phi^+(x)\phi^+(y) + \phi^-(x)\phi^-(y) + \phi^-(x)\phi^+(y) + \phi^-(y)\phi^+(x) + [\phi^+(x), \phi^-(y)]$$

therefore, we have :

$$\phi(x)\phi(y) = \phi^+(x)\phi^+(y) + \phi^-(x)\phi^-(y) + \phi^-(x)\phi^+(y) + \phi^-(y)\phi^+(x) + [\phi^+(x), \phi^-(y)] \tag{4.16}$$

In every term except commutator ,all $a_{\vec{p}}$ are to the right of all $a_{\vec{p}}$,have normal ordered arrangement. All such normal ordered operators have zero vacuum expectation value.

For $y^0 > x^0$:

$$\phi(y)\phi(x) = [\phi^+(y) + \phi^-(y)][\phi^+(x) + \phi^-(x)]$$

$$= \phi^+(y)\phi^+(x) + \phi^+(y)\phi^-(x) + \phi^-(y)\phi^+(x) + \phi^-(y)\phi^-(x)$$

using commutator as before, we can write it as:

$$\phi(y)\phi(x) = \phi^+(y)\phi^+(x) + \phi^-(x)\phi^-(y) + \phi^-(y)\phi^+(x) + \phi^-(x)\phi^+(y) + [\phi^+(y), \phi^-(x)] \tag{4.17}$$

Contraction

The **contraction** is defined by :

$$\overbracket{\phi(x)\phi}(y) = \begin{cases} [\phi^+(x), \phi^-(y)] & \text{for } x^0 > y^0 \\ [\phi^+(y), \phi^-(x)] & \text{for } y^0 > x^0 \end{cases} \tag{4.18}$$

This is actually the Feynman propagator ,

$$\overbracket{\phi(x)\phi}(y) = G_F(x - y) \tag{4.19}$$

Now ,using equations (4.16) and (4.17) , we can write :

$$T[\phi(x)\phi(y)] = N[\phi(x)\phi(y) + \overbracket{\phi(x)\phi}(y)]$$

where N is the normal ordering symbol.

In general, for 'm' number of fields, we have ;

$$\boxed{T[\phi(x_1)\phi(x_2)\cdots\phi(x_m)] = N\left[\phi(x_1)\phi(x_2)\cdots\phi(x_m) + \text{all possible contractions}\right]} \tag{4.19}$$

This way of expanding products of fields in terms of normal ordering and contractions ,is known as the **Wick's theorem**.

4.3 S-matrix

At $t \to -\infty$ and $t \to +\infty$, the incoming and outgoing particles in scattering process are described by the free particle states which are called **asymptotic states**.

The initial state of the incoming particle is given by

$$|\psi_i(-\infty)\rangle = |i\rangle$$

then

$$|\psi_i(+\infty)\rangle = U(+\infty,-\infty)|\psi_i(-\infty)\rangle = U(+\infty,-\infty)|i\rangle = S|i\rangle$$

where

$$S = U(+\infty,-\infty) \tag{4.20}$$

is called the **S-matrix** (S stands for scattering).

The probability amplitude or the scattering amplitude is defined by :

$$S_{fi} = \langle f|\psi_i(+\infty)\rangle = \langle f|U(+\infty,-\infty)|i\rangle$$

We know that

$$U(t,t_0) = T\left[\exp\left(-i\int_{t_0}^{t} H_I(t')dt'\right)\right]$$

therefore, using equation (4.20),we have

$$S = U(+\infty,-\infty) = T\left[\exp\left(-i\int_{-\infty}^{+\infty} H_I(t')dt'\right)\right] \tag{4.21}$$

Using the Dyson series for U ,we can write S-matrix as :

$$\begin{aligned} S &= \sum_{n=0}^{\infty}\frac{(-i)^n}{n!}\int_{-\infty}^{+\infty}dt_1\int_{-\infty}^{+\infty}dt_2\cdots\int_{-\infty}^{+\infty}dt_n\, T[H_I(t_1)H_I(t_2)\cdots H_I(t_n)] \\ &= 1 - i\int_{-\infty}^{+\infty}dt\, H_I(t) + \frac{(-i)^2}{2!}\int_{-\infty}^{+\infty}dt_1\int_{-\infty}^{+\infty}dt_2\, T[H_I(t_1)H_I(t_2)] + \cdots \end{aligned} \tag{4.22}$$

This is written as

$$S = 1 + iT \tag{4.23}$$

to get the scattering amplitude and since $\langle f|1|i\rangle$ is trivially non-scattering term ,we write

$$\langle f|(S-1)|i\rangle = \langle f|iT|i\rangle \tag{2.24}$$

The two main important properties of s-matrix are :

1. **Unitary :** $S^\dagger S = SS^\dagger = \mathbb{1}$
2. **Lorentz Invariance:** The S-matrix is invariant under the Lorentz transformation.

4.4 Yukawa Interactions Theory

This theory explains the interaction between the Dirac field and a neutral scalar (Klein–Gordon) field. The interaction Lagrangian can be expressed as:

$$\mathcal{L} = \bar{\psi}(i\gamma^\mu\partial_\mu - m)\psi + \frac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) - \frac{1}{2}M^2\phi^2 - g\,\bar{\psi}\psi\,\phi \tag{4.25}$$

where m is the mass of Dirac field's particle and M is the mass of scalar field's particle and g is the coupling constant.

Now the total Lagrangian can be written as :

$$\mathcal{L} = \mathcal{L}_0 + \mathcal{L}_{\text{int}}$$

where ;

$$\mathcal{L}_0 = \bar{\psi}(i\gamma^\mu\partial_\mu - m)\psi + \frac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) - \frac{1}{2}M^2\phi^2$$

is the **free Lagrangian** for Dirac and scalar fields and

$$\mathcal{L}_{\text{int}} = -g\,\bar{\psi}\psi\,\phi \tag{4.26}$$

is the **interaction Lagrangian**.

So now we have $\mathcal{L}_\mathcal{I} = \mathcal{H}_\mathcal{I} = -g\,\bar{\psi}\psi\,\phi$ therefore,

$$H_I = \int d^3\vec{x}\mathcal{H}_\mathcal{I}$$

using thing in equation (4.21) ,we get :

$$
\begin{aligned}
S &= T\exp\left[-i\int_{-\infty}^{\infty} dt\, H_I(t)\right] \\
&= T\exp\left[-i\int d^4x\, \mathcal{L}_I(x)\right] \\
&= T\exp\left[-ig\int d^4x\, \bar{\psi}(x)\psi(x)\phi(x)\right]
\end{aligned} \tag{4.27}
$$

We can expand this as :

$$
\begin{aligned}
S &= \sum_{n=0}^{\infty}\frac{1}{n!}\left[-ig\int d^4x\, \bar{\psi}(x)\psi(x)\phi(x)\right]^n \\
&= 1 + (-ig)\int d^4x_1\, T[\bar{\psi}(x_1)\psi(x_1)\phi(x_1)] \\
&\quad + \frac{(-ig)^2}{2!}\int d^4x_1 d^4x_2\, T[\bar{\psi}(x_1)\psi(x_1)\phi(x_1)\bar{\psi}(x_2)\psi(x_2)\phi(x_2)] + \cdots
\end{aligned} \tag{4.28}
$$

4.4.1 Derivation of 2nd Order S-matrix Element

For the 2nd order expansion ,we can write using equation (4.28) as :

$$
S^{(2)} = \frac{(-ig)^2}{2!}\int d^4x_1 d^4x_2\, T[\bar{\psi}(x_1)\psi(x_1)\phi(x_1)\bar{\psi}(x_2)\psi(x_2)\phi(x_2)] \tag{4.29}
$$

By using the Wick's theorem (4.19), we can expand the time ordered componet as :

$$
\begin{aligned}
&T[\bar{\psi}(x_1)\psi(x_1)\phi(x_1)\bar{\psi}(x_2)\psi(x_2)\phi(x_2)] \\
&=: \bar{\psi}(x_1)\psi(x_1)\phi(x_1)\bar{\psi}(x_2)\psi(x_2)\phi(x_2) : \\
&\quad + : \bar{\psi}(x_1)\psi(x_1)\bar{\psi}(x_2)\psi(x_2) : \overbracket{\phi(x_1)\phi}(x_2) \\
&\quad + \text{other possible contraction terms}
\end{aligned} \tag{4.30}
$$

I have written only relevant terms. Only the 2nd term contributes non-zero value to the scattering-matrix ,rest of the terms give zero values upon taking expectation value in the scattering states.

4.4.2 Nucleon-Nucleon Scattering

The nucleon-nucleon scattering is reprented by $\psi\psi \to \psi\psi$ scattering.The nucleons are fermion or Dirac particles having non-zero spins. **But for convenience of calculations in this section,we consider nucleons to be spinless particles**. The initial and final

states are given by

$$
\begin{aligned}
|i\rangle &= \sqrt{4E_{\vec{p}_1}E_{\vec{p}_2}}\, b^{\dagger}_{\vec{p}_1} b^{\dagger}_{\vec{p}_2}|0\rangle = |\vec{p}_1, \vec{p}_2\rangle \equiv |\vec{p}_1, \vec{p}_2\rangle \\
|f\rangle &= \sqrt{4E_{\vec{p}'_1}E_{\vec{p}'_2}}\, b^{\dagger}_{\vec{p}'_1} b^{\dagger}_{\vec{p}'_2}|0\rangle = |\vec{p}'_1, s'_1; \vec{p}'_2, s'_2\rangle \equiv |\vec{p}'_1, \vec{p}'_2\rangle
\end{aligned} \tag{4.31}
$$

Also , the adjoint is given by :

$$
\langle f| = \sqrt{4E_{\vec{p}'_1}E_{\vec{p}'_2}}\,\langle 0| b_{\vec{p}'_2} b_{\vec{p}'_1} \tag{4.32}
$$

Now by using equations (4.29), (4.30) ,(4.31) and (4.32) , we can write the 2nd order scattering amplitude as :

$$
\begin{aligned}
S^{(2)}_{fi} &= \langle f|S^{(2)}|i\rangle \\
&= \frac{(-ig)^2}{2!}\int d^4x_1 d^4x_2\, \langle p'_1, p'_2| : \bar{\psi}(x_1)\psi(x_1)\bar{\psi}(x_2)\psi(x_2) : \overbracket{\phi(x_1)\phi}(x_2)|p_1, p_2\rangle
\end{aligned} \tag{4.33}
$$

or

$$
\mathcal{M} = \frac{(-ig)^2}{2!}\int d^4x_1 d^4x_2\, G_F(x_1 - x_2)\langle p'_1, p'_2| : \bar{\psi}(x_1)\psi(x_1)\bar{\psi}(x_2)\psi(x_2) : |p_1, p_2\rangle \tag{4.34}
$$

where

$$
\begin{aligned}
G_F(x_1 - x_2) &= \overbracket{\phi(x_1)\phi}(x_2) = \langle 0|T[\phi(x_1)\phi(x_2)]|0\rangle \\
&= \int \frac{d^4p}{(2\pi)^4}\frac{i}{p^2 - m^2 + i\epsilon}e^{ip\cdot(x_1 - x_2)}
\end{aligned} \tag{4.35}
$$

is the **Feynman propagator** for the scalar fields $\phi(x_1)$ and $\phi(x_2)$.

Now we write using property of normal ordering in which create operator is placed in left of the annihilation operator and since $\gamma^0\gamma^0 = 1$;

$$
\langle p'_1, p'_2| : \psi^{\dagger}(x_1)\psi(x_1)\psi^{\dagger}(x_2)\psi(x_2) : |p_1, p_2\rangle = \langle p'_1, p'_2|\psi^{\dagger}(x_1)\psi^{\dagger}(x_2)\psi(x_1)\psi(x_2)|p_1, p_2\rangle \tag{4.36}
$$

We know that

$$
\langle 0|\psi(x)|p\rangle = e^{-ip\cdot x}
$$

and

$$
\langle p|\psi^{\dagger}(x)|0\rangle = e^{ip\cdot x}
$$

using these ,we can equation (4.36) as :

$$\begin{aligned}
&\langle p_1', p_2'| : \psi^\dagger(x_1)\psi(x_1)\psi^\dagger(x_2)\psi(x_2) : |p_1, p_2\rangle \\
&= \langle p_1', p_2'|\psi^\dagger(x_1)\psi^\dagger(x_2)|0\rangle\langle 0|\psi(x_1)\psi(x_2)|p_1, p_2\rangle \\
&= \left(e^{ip_1'\cdot x_1 + ip_2'\cdot x_2} + e^{ip_1'\cdot x_2 + ip_2'\cdot x_1}\right) \\
&\quad \times \left(e^{-ip_1\cdot x_1 - ip_2\cdot x_2} + e^{-ip_1\cdot x_2 - ip_2\cdot x_1}\right) \\
&= 2\left(e^{ix_1\cdot(p_1'-p_1)+ix_2\cdot(p_2'-p_2)} + e^{ix_1\cdot(p_2'-p_1)+ix_2\cdot(p_1'-p_2)}\right)
\end{aligned} \tag{4.37}$$

Substituting equations (4.35) and (4.37) in (4.34), we get :

$$\begin{aligned}
\mathcal{M} = (-ig)^2 \int \frac{d^4p}{(2\pi)^4} \frac{i}{p^2 - m^2 + i\epsilon} \int d^4x_1 d^4x_2\, e^{ip\cdot(x_1-x_2)} \\
\times \left(e^{ix_1\cdot(p_1'-p_1)+ix_2\cdot(p_2'-p_2)} + e^{ix_1\cdot(p_2'-p_1)+ix_2\cdot(p_1'-p_2)}\right)
\end{aligned} \tag{4.38}$$

Integrating over x_1 and x_2, we obtain ;

$$\begin{aligned}
\mathcal{M} = (-ig)^2 \int \frac{d^4p}{(2\pi)^4} \frac{i}{p^2 - m^2 + i\epsilon} (2\pi)^4 (2\pi)^4 \\
\times \Big[\delta^4(p + p_1' - p_1)\delta^4(-p + p_2' - p_2) \\
+ \delta^4(p + p_2' - p_1)\delta^4(-p + p_1' - p_2)\Big]
\end{aligned} \tag{4.1}$$

further integrating over p;

$$\begin{aligned}
\mathcal{M} = (-ig)^2 (2\pi)^4 \Big[\frac{i}{(p_1 - p_1')^2 - m^2 + i\epsilon} \delta^4(p_1 + p_2 - p_1' - p_2') \\
+ \frac{i}{(p_1 - p_2')^2 - m^2 + i\epsilon} \delta^4(p_1 + p_2 - p_1' - p_2')\Big]
\end{aligned} \tag{4.2}$$

or

$$\begin{aligned}
\mathcal{M} = i(-ig)^2 (2\pi)^4 \delta^4(p_1 + p_2 - p_1' - p_2') \\
\times \Big[\frac{1}{(p_1 - p_1')^2 - m^2 + i\epsilon} + \frac{1}{(p_1 - p_2')^2 - m^2 + i\epsilon}\Big]
\end{aligned} \tag{4.38}$$

4.4.3 Amplitude in the Centre Of Mass Frame

In the centre of mass frame , $\vec{p}_1 = -\vec{p}_2$, and by momentum conservation , $|\vec{p}_1| = |\vec{p}_1'|$, also we can drop the $i\epsilon$ since the denominator is never zero ,therefore,in the COM

frame, the amplitude from equation (3.38) can be write :

$$\begin{aligned}\mathcal{M} = i(-ig)^2(2\pi)^4\delta^4(p_1 + p_2 - p_1' - p_2') \\ \times \Big[\frac{1}{(p_1 - p_1')^2 - m^2} + \frac{1}{(p_1 - p_2')^2 - m^2}\Big]\end{aligned} \tag{4.39}$$

This amplitude is equally applicable to other scattering processes, for instance, $\bar{\psi}\bar{\psi} \to \bar{\psi}\bar{\psi}$ and $\psi\bar{\psi} \to \psi\psi$.

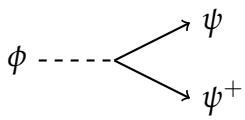
Thus,the same amplitude can represent distinct physical scattering processes and this is the important feature to be used in the Feynman diagrams.

4.5 Feynman Diagram And Feynman Rules for Spinless Particles

The Feynman diagram is used to represent the perturbative expansion of $\langle f|S|i\rangle$ under the Wick's theorem and so it helps us to calculate the amplitudes in a systematic and convenient way.

The different terms arising from Wick's expansion can be visualized graphically as follows:

- Represent each particle in the initial state $|i\rangle$ and final state $|f\rangle$ with an external line.
- Use dotted lines to denote mesons and solid lines for fermions, assigning a momentum p to each line.
- Indicate the charge flow by adding arrows along the solid lines.
- Connect the external lines at a common point to form a trivalent vertex.



Each of these diagrams corresponds uniquely to a specific term appearing in the expansion of $\langle f|S - 1|i\rangle$.

4.5.1 Feynman Rules

- Assign a momentum k to every internal line in the diagram.

- For each vertex, include a factor

$$(-ig)(2\pi)^4\delta^{(4)}\left(\sum_i k_i\right),$$

  where $\sum_i k_i$ represents the total momentum entering that vertex.
- For every internal dotted line corresponding to a ϕ particle carrying momentum k, associate the following factor:

$$\int \frac{d^4k}{(2\pi)^4}\frac{i}{k^2 - m^2 + i\epsilon}$$

We assign an identical factor to the internal solid lines representing the ψ particles, with m replaced by the corresponding fermion mass M.

4.5.2 Example of Nucleon-Nucleon scattering

We again the example of nucleon-nucleon scattering as discussed in subsection (4.4.2) ;

$$\psi\psi \to \psi\psi$$

This scattering process of spinless nucleons can be given by the two Feynman diagrams :

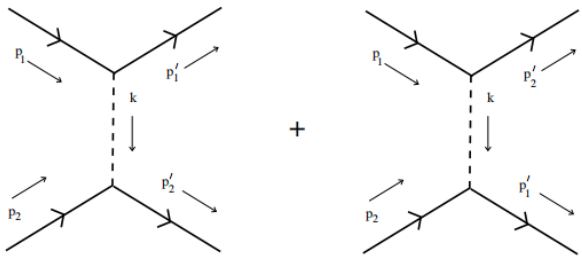


Figure 4.1: The two lowest 2nd order Feynman diagrams for spinless nucleon scattering

A scalar meson ϕ of mass m is exchanged between two spinless nucleons of mass M.

Feynman Rules To be Used

We apply the Feynman rules introduced in subsection (4.5.1) in a simplified manner as outlined below:

- Assign a factor of $(-ig)$ to every interaction vertex.
- For each internal line, use the propagator corresponding to the scalar meson.
- Perform an integration over the loop momentum k according to

$$\int \frac{d^4k}{(2\pi)^4}.$$

For the first left diagram in figure 4.1 ;

$$\mathcal{M}_1 \;=\; \int \frac{d^4k}{(2\pi)^4}\,(-ig)(-ig)\,\frac{i}{k^2-m^2+i\epsilon}\,(2\pi)^4\delta^{(4)}(p_1-p_1'-k)\,(2\pi)^4\delta^{(4)}(p_2-p_2'+k)$$

Integrating over k by using the delta functions (which sets $k = p_1 - p_1'$) ,we get :

$$\mathcal{M}_1 = (-ig)^2\,\frac{i}{(p_1-p_1')^2-m^2}\,(2\pi)^4\delta^{(4)}(p_1+p_2-p_1'-p_2'),$$

for the right diagram of figure 4.1 , we have

$$\mathcal{M}_2 \;=\; \int \frac{d^4k}{(2\pi)^4}\,(-ig)(-ig)\,\frac{i}{k^2-m^2+i\epsilon}\,(2\pi)^4\delta^{(4)}(p_1-p_2'-k)\,(2\pi)^4\delta^{(4)}(p_2-p_1'+k)$$

integrating over k ,we get ;

$$\mathcal{M}_2 = (-ig)^2\frac{i}{(p_1-p_2')^2-m^2}\,(2\pi)^4\delta^{(4)}(p_1+p_2-p_1'-p_2').$$

Therefore, the total amplitude for the two diagrams is :

$$\begin{aligned}\mathcal{M} &= \mathcal{M}_1+\mathcal{M}_2\\ &= i(-ig)^2(2\pi)^4\delta^{(4)}(p_1+p_2-p_1'-p_2')\left[\frac{1}{(p_1-p_1')^2-m^2+i\epsilon}+\frac{1}{(p_1-p_2')^2-m^2+i\epsilon}\right].\end{aligned} \tag{4.40}$$

This amplitude is exactly the same as equation (4.39) derived using Wick's theorem with lots of calculations and thus , the Feynman diagram simplifies the tedious

calculations in a simple approach.

Chapter–4 Summary

In this last chapter, we created the framework for interacting quantum fields. We derived the Dyson series and formulated Wick's theorem for time-ordered operator products. The S-matrix, defined as $S = U(+\infty, -\infty)$, was introduced to describe scattering phenomena.

In the context of Yukawa theory, we calculated the second-order scattering amplitude $\mathcal{M}$ and set up the corresponding Feynman rules for processes involving spinless particle interactions. This formulation lays the groundwork for the diagrammatic evaluation of scattering amplitudes in perturbative quantum field theory.

Conclusion

- The Quantum Field Theory combines the ideas of special relativity and quantum mechanics. This report began with the basic formulation of field theory. It developed Lagrangian and Hamiltonian formalisms, showing the important connection between continuous symmetries and corresponding conservation laws as established by Noether's theorem. These foundations help us understand how physical observables come from the dynamics of fields.

- The canonical quantization of the real scalar (Klein–Gordon) field revealed that it describes spin-zero(spinless) particles. During this process, the structure of Fock space emerged, introducing creation and annihilation operators that define multiparticle states. The Feynman propagator appeared as the causal Green's function, highlighting its essential role in mediating interactions and determining transition amplitudes in relativistic quantum theories.

- When extending the formalism to spin-$\frac{1}{2}$ fields, the Dirac equation provided a consistent quantum description of fermions and predicted the existence of antiparticles. The interaction picture, Dyson series, and Wick's theorem were used to handle interacting theories and calculate scattering amplitudes. The Yukawa model illustrated how Feynman diagrams turn complex operator algebra into clear, calculable representations of physical processes.

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